Jerrold E. Marsden and Anthony J. Tromba

Vector Calculus

Fifth Edition

Chapter 8:
The Integral Theorems of Vector Analysis

8.1 Green’s Theorem

8.1 Green's Theorem

Key Points in this Section.

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1. **Statement of Green's Theorem.** For a simple region D with bounding curve $C = \partial D$ and two C^1 functions P and Q on D, we have
$$\int_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx\,dy$$
2. **Orientation.** The orientation is chosen so that as you proceed along the boundary curve in the positive direction, the region is on your left. For simple regions this means that you go around the regions *counter-clockwise*; if there are holes inside the region, those boundaries get traversed *clockwise*.

3. **Strategy of the Proof.** For a y-simple region, one proves by reduction to iterated integrals, the Fundamental Theorem of Calculus and the definition of the line integral that

$$\int_C P\,dx = -\iint_D \left(\frac{\partial P}{\partial y}\right) dx\,dy$$

Similarly, for a x-simple region, we have

$$\int_C Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x}\right) dx\,dy$$

One gets Green's theorem for simple regions by simply adding these two results.

4. **More General Regions.** One gets Green's theorem for more general regions by breaking up a given region into simple ones as in Figure 8.1.5 of the Text. Here is another example of how to break up a region.

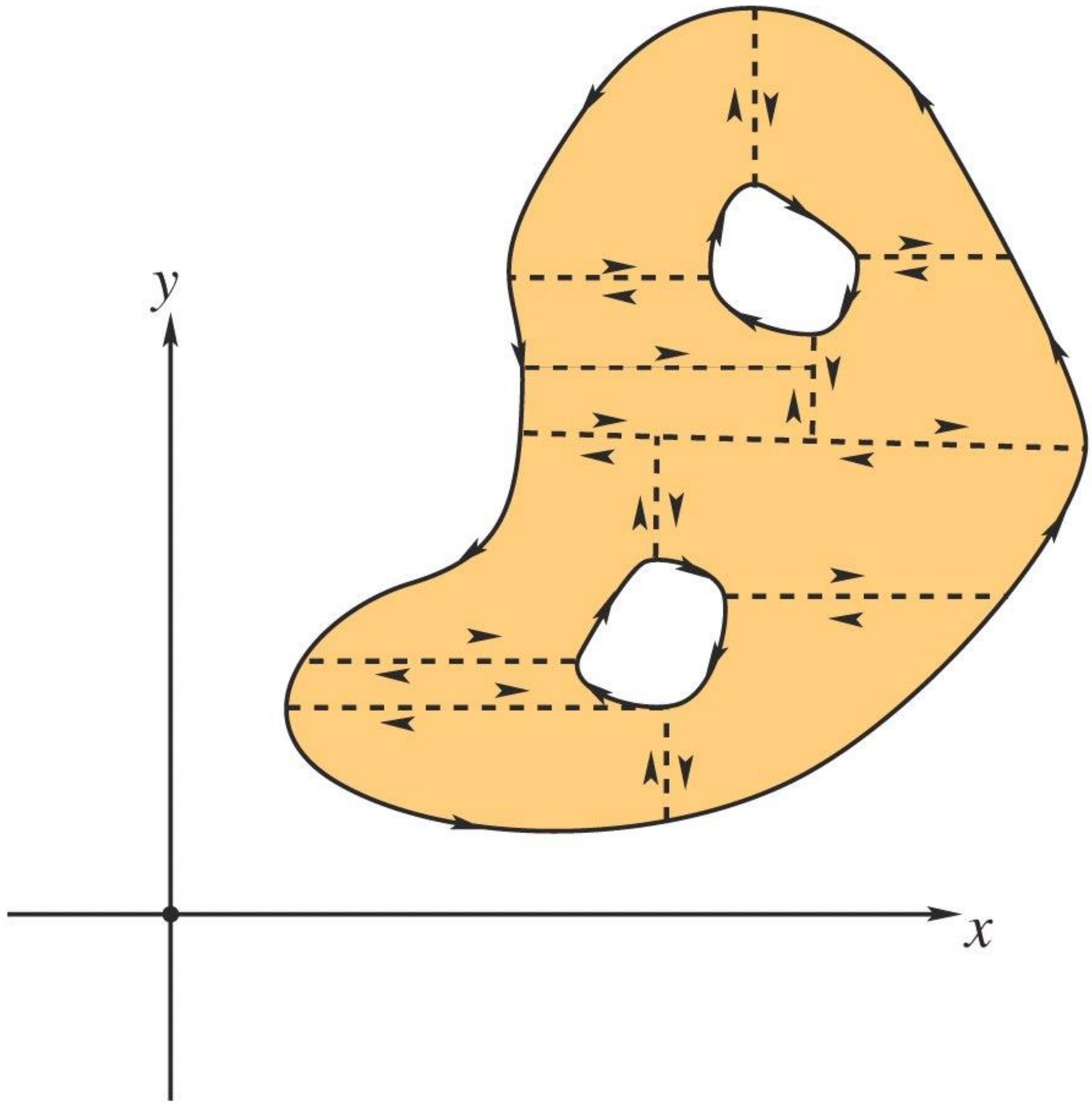


FIGURE 8.1.1. How to break a two-holed region up into simple regions.

5. **Area.** As a special case of Green's theorem, one finds that the area of a region is

$$A = \frac{1}{2}\int_{\partial D} x\,dy - y\,dx$$

6. **Vector form of Green's theorem.** If $\mathbf{F}$ is a vector field in the plane, then

$$\int_{\partial D} \mathbf{F}\cdot d\mathbf{s} = \iint_D (\nabla\times\mathbf{F})\cdot\mathbf{k}\,dx\,dy.$$

   This is proved by simply writing $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ and applying Green's theorem and noting that

$$\nabla\times\mathbf{F} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathbf{k}.$$

7. **Divergence theorem in the plane.** This result says that

$$\int_{\partial D} \mathbf{F} \cdot \mathbf{n}\, ds = \iint_D (\operatorname{div} \mathbf{F})\, dx\, dy.$$

where $\mathbf{n}$ is the outward normal to the boundary. This is proved by again writing $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ and noting that the unit outward normal is given by

$$\mathbf{n} = \frac{y'\mathbf{i} - x'\mathbf{j}}{\sqrt{(x')^2 + (y')^2}}$$

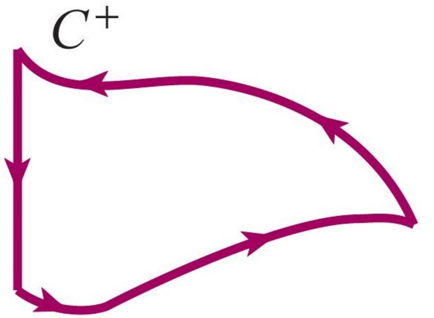
using

$$ds = \sqrt{(x')^2 + (y')^2}\, dt,$$

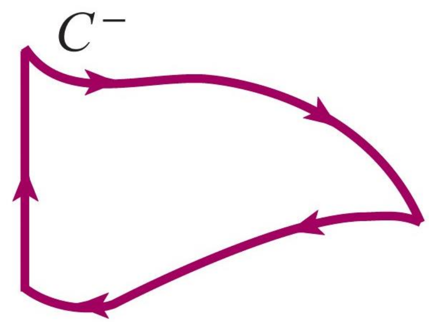
substituting into the left side to get

$$\int_{\partial D} P\, dy - Q\, dx,$$

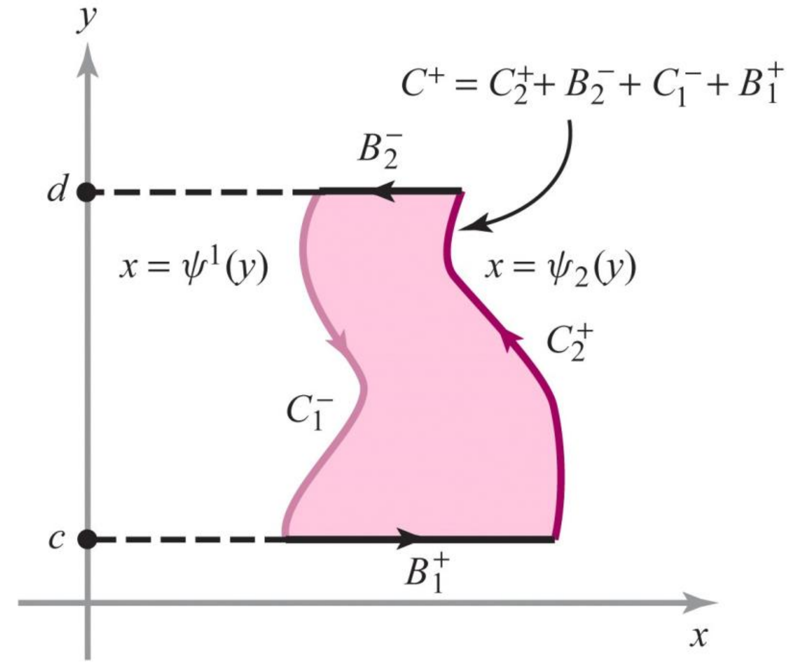
and then using Green's theorem.



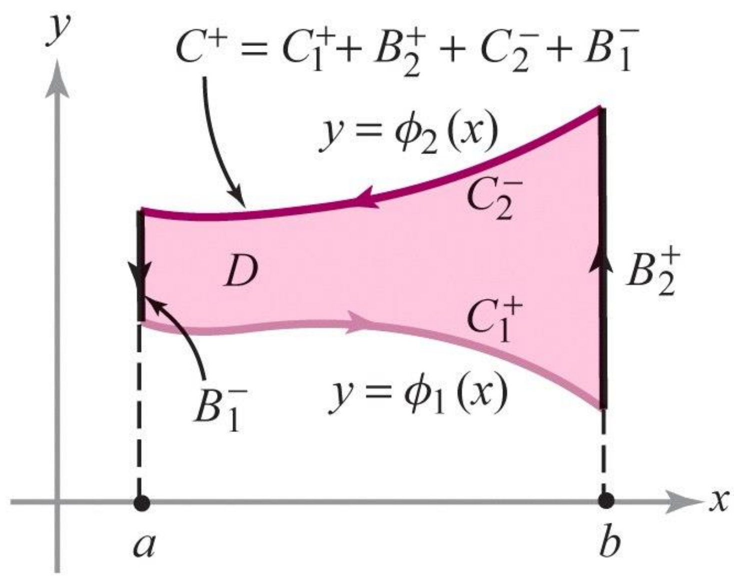
Positive orientation

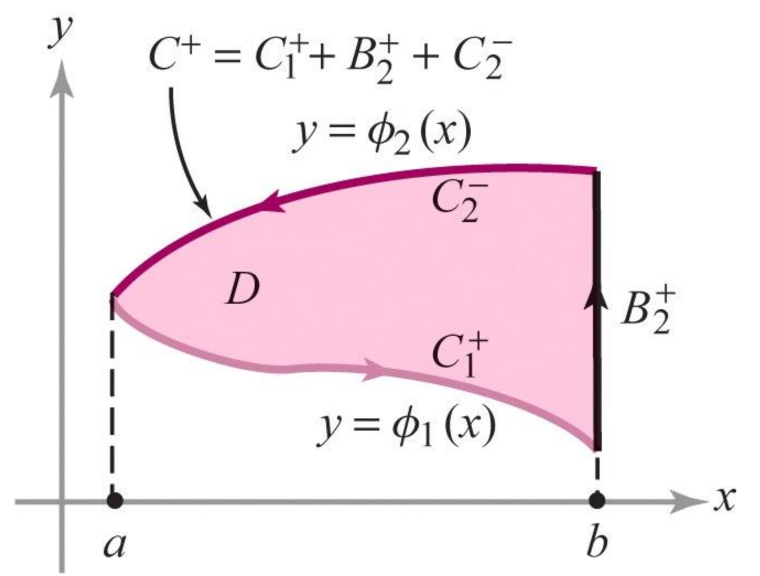


Negative orientation



Si camines per la vora $C = \partial D$ en orientació positiva, C^+, la regió D queda sempre a l'esquerra





THEOREM 1: Green's Theorem Let D be a simple region and let C be its boundary. Suppose $P: D \to \mathbb{R}$ and $Q: D \to \mathbb{R}$ are of class C^1. Then

$$\int_{C^+} P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy.$$

$$\boldsymbol{F}(x,y) = P(x,y)\boldsymbol{i} + Q(x,y)\boldsymbol{j}$$

$$\int_{C^+} \boldsymbol{F} \cdot d\boldsymbol{s} = \int_{C^+} P\,dx + Q\,dy$$

Proof

$$\int_{C^+} P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx\,dy$$

Based on proving separately the following two Lemmas

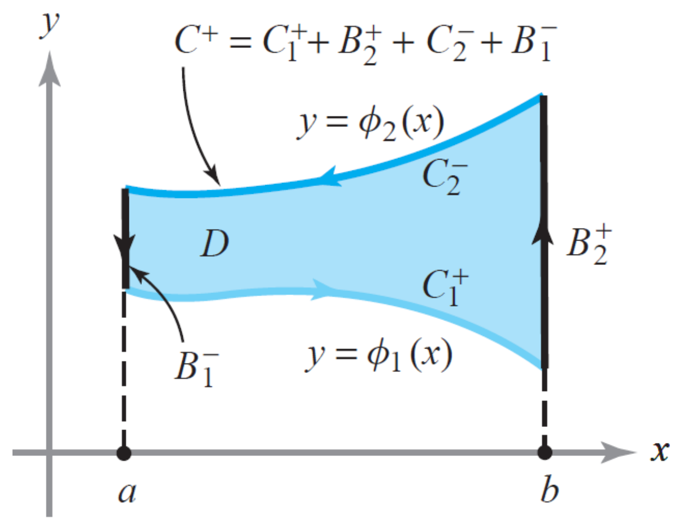
Lemma 1 Let D be a y-simple region and let C be its boundary. Suppose $P\colon D \to \mathbb{R}$ is of class C^1. Then

$$\int_{C^+} P\,dx = -\iint_D \frac{\partial P}{\partial y}\,dx\,dy.$$

Lemma 2 Let D be an x-simple region with boundary C. Then if $Q\colon D \to \mathbb{R}$ is C^1,

$$\int_{C^+} Q\,dy = \iint_D \frac{\partial Q}{\partial x}\,dx\,dy.$$

Proof of Lemma 1



By Fubinis's Theorem and Fundamental Theorem of Calculus

$$\iint_D \frac{\partial P}{\partial y}(x, y)\, dx\, dy = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} \frac{\partial P}{\partial y}(x, y)\, dy\, dx$$

$$= \int_a^b [P(x, \phi_2(x)) - P(x, \phi_1(x))]\, dx.$$

Parametrizing C_1^+ as $x \to (x, \phi_1(x))$ for $a \le x \le b$, and C_2^+ as $x \to (x, \phi_2(x))$ for $a \le x \le b$,

$$\int_a^b P(x, \phi_1(x))\, dx = \int_{C_1^+} P(x, y)\, dx$$

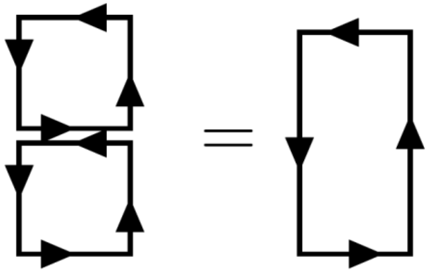
$$\int_a^b P(x, \phi_2(x))\, dx = \int_{C_2^+} P(x, y)\, dx = -\int_{C_2^-} P(x, y)\, dx.$$

The integrals on B_1^- and B_2^+ are zero because $\boldsymbol{F} = P\boldsymbol{i}$ and $d\boldsymbol{s} \perp \boldsymbol{F}$, thus

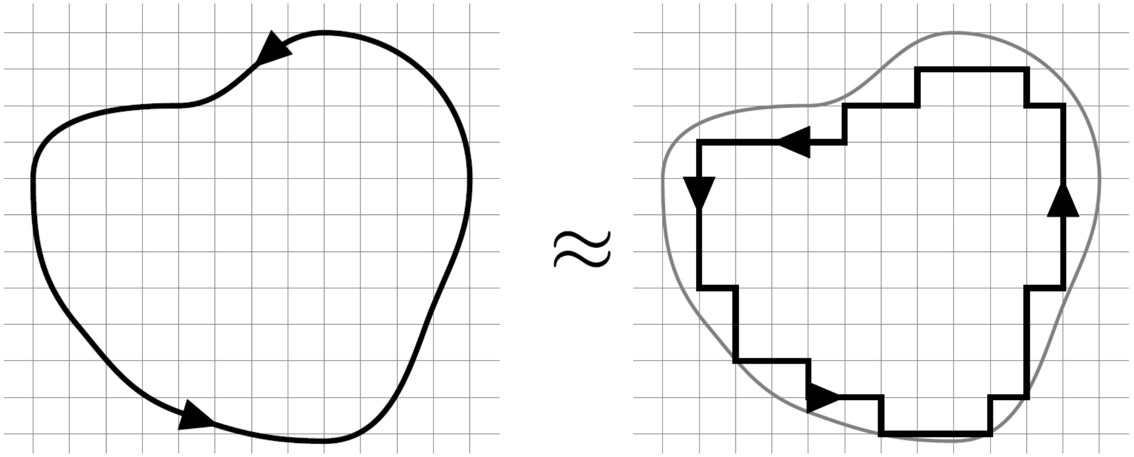
$$\iint_D \frac{\partial P}{\partial y}\, dx\, dy = -\int_{C_1^+} P\, dx - \int_{C_2^-} P\, dx = -\int_{C^+} P\, dx$$

Another approach to proving Green's theorem

1) Show the theorem applies to rectangles (as before)
2) Realize that the line integrals of two adjacent rectangles is equal to the line integral of the joined rectangle



3) The same applies to the surface integrals (but is trivial in this case
4) Decompose the region using rectangles

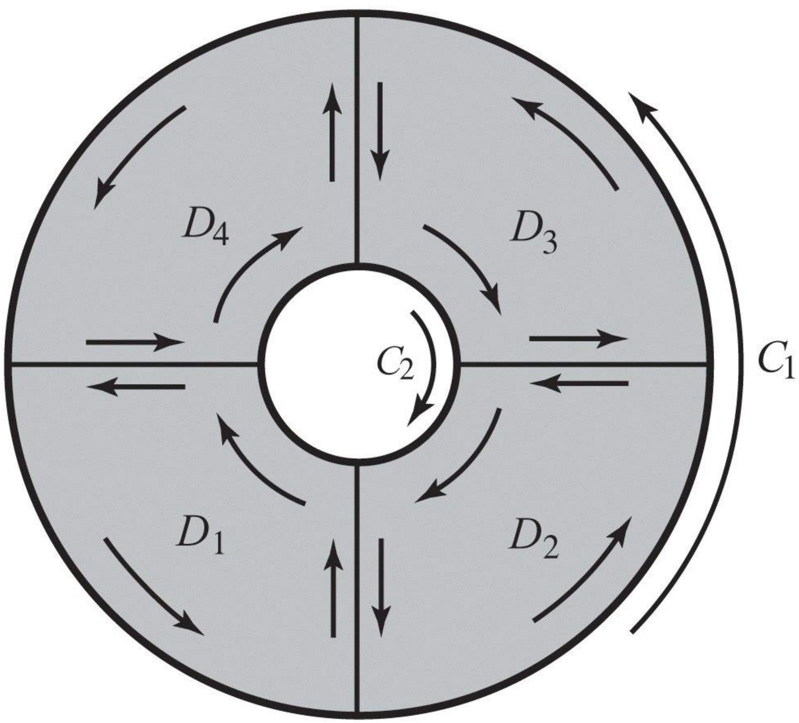


5) Take care of the limit on the boundary

Generalizing Green´s Theorem

També serveix si la regió no és simple, per descomposició en unió de regions simples

$$\int_{\partial D} P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx\,dy.$$



$D = D_1 \cup D_2 \cup D_3 \cup D_4$

THEOREM 2: Area of a Region If C is a simple closed curve that bounds a region to which Green's theorem applies, then the area of the region D bounded by $C = \partial D$ is

$$A = \frac{1}{2} \int_{\partial D} x\,dy - y\,dx.$$

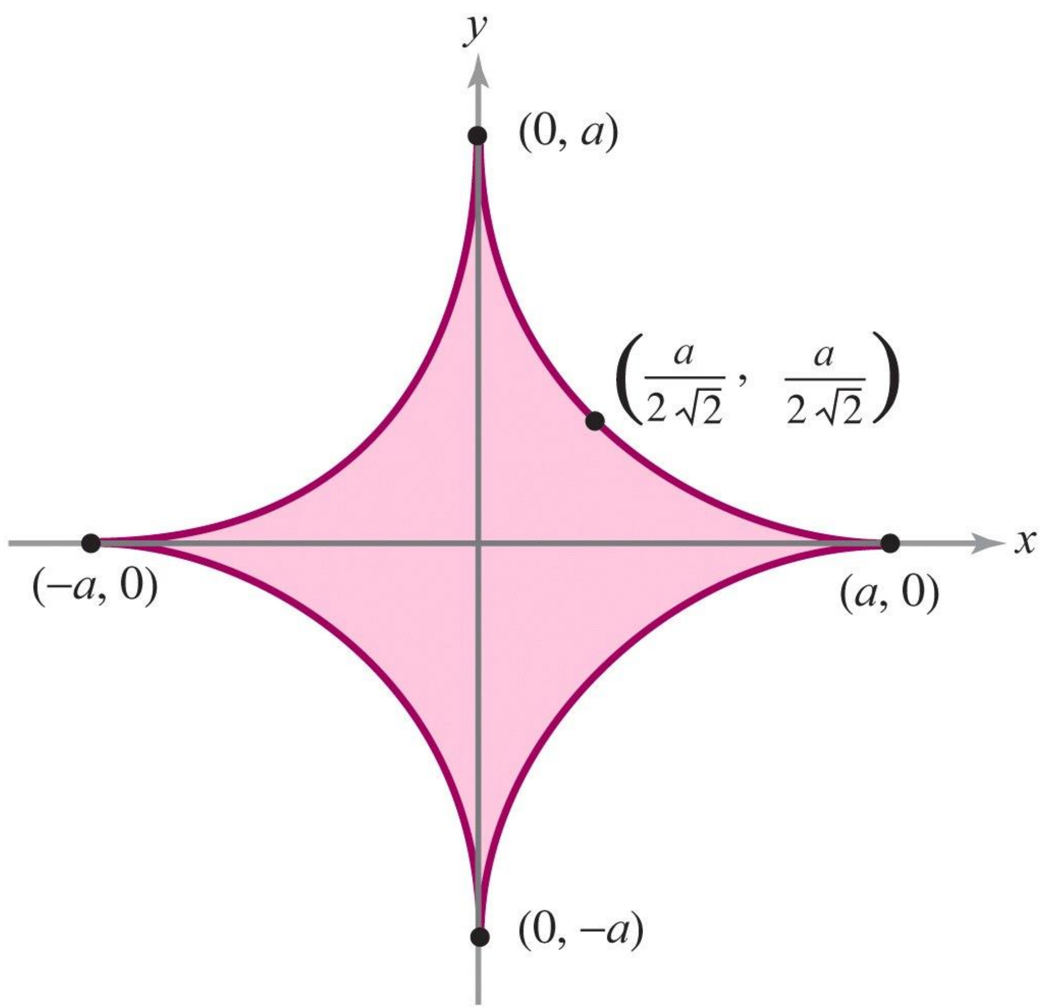
proof Let $P(x, y) = -y$, $Q(x, y) = x$; then by Green's theorem we have

$$\begin{aligned} \frac{1}{2} \int_{\partial D} x\,dy - y\,dx &= \frac{1}{2} \iint_D \left[\frac{\partial x}{\partial x} - \frac{\partial(-y)}{\partial y} \right] dx\,dy \\ &= \frac{1}{2} \iint_D [1 + 1]\,dx\,dy = \iint_D dx\,dy = A. \end{aligned}$$

■

Let $a > 0$. Compute the area (see Figure 8.1.6) of the region enclosed by the hypocycloid defined by $x^{2/3} + y^{2/3} = a^{2/3}$ using the parametrization

$$x = a\cos^3\theta, \qquad y = a\sin^3\theta, \qquad 0 \le \theta \le 2\pi.$$



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$$x = a\cos^3\theta, \qquad y = a\sin^3\theta, \qquad 0 \le \theta \le 2\pi.$$

$$\begin{aligned}
A &= \frac{1}{2}\int_{\partial D} x\,dy - y\,dx \\
&= \frac{1}{2}\int_0^{2\pi} [(a\cos^3\theta)(3a\sin^2\theta\cos\theta) - (a\sin^3\theta)(-3a\cos^2\theta\sin\theta)]\,d\theta \\
&= \frac{3}{2}a^2\int_0^{2\pi} (\sin^2\theta\cos^4\theta + \cos^2\theta\sin^4\theta)\,d\theta = \frac{3}{2}a^2\int_0^{2\pi} \sin^2\theta\cos^2\theta\,d\theta \\
&= \frac{3}{8}a^2\int_0^{2\pi} \sin^2 2\theta\,d\theta = \frac{3}{8}a^2\int_0^{2\pi}\left(\frac{1-\cos 4\theta}{2}\right)d\theta \\
&= \frac{3}{16}a^2\int_0^{2\pi} d\theta - \frac{3}{16}a^2\int_0^{2\pi}\cos 4\theta\,d\theta = \frac{3}{8}\pi a^2.
\end{aligned}$$

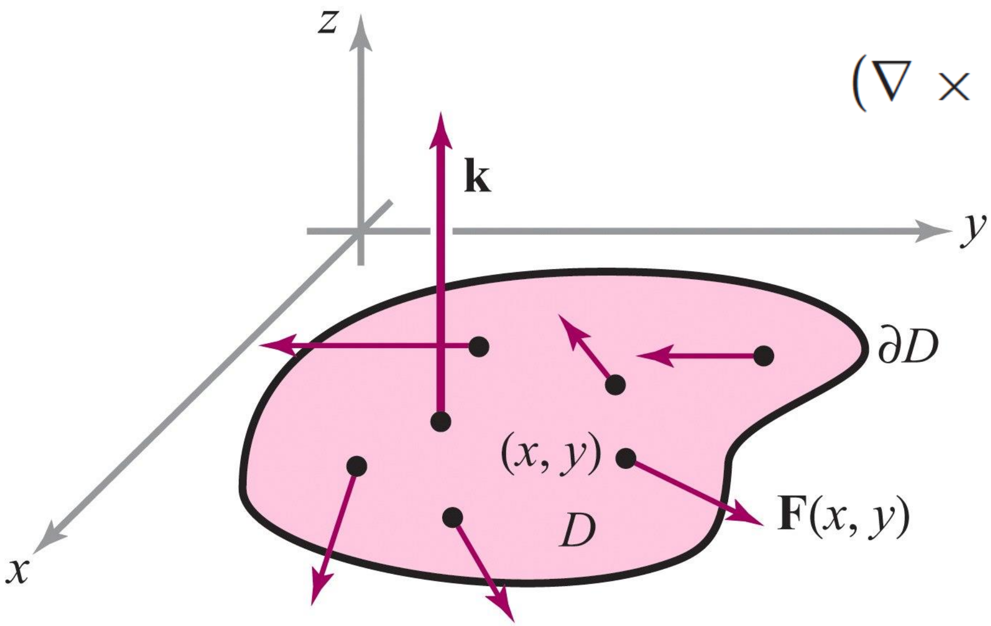
▲

Teorema de Green utilitzant el rotacional

THEOREM 3: Vector Form of Green's Theorem Let $D \subset \mathbb{R}^2$ be a region to which Green's theorem applies, let ∂D be its boundary (oriented counterclockwise), and let $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ be a C^1 vector field on D. Then

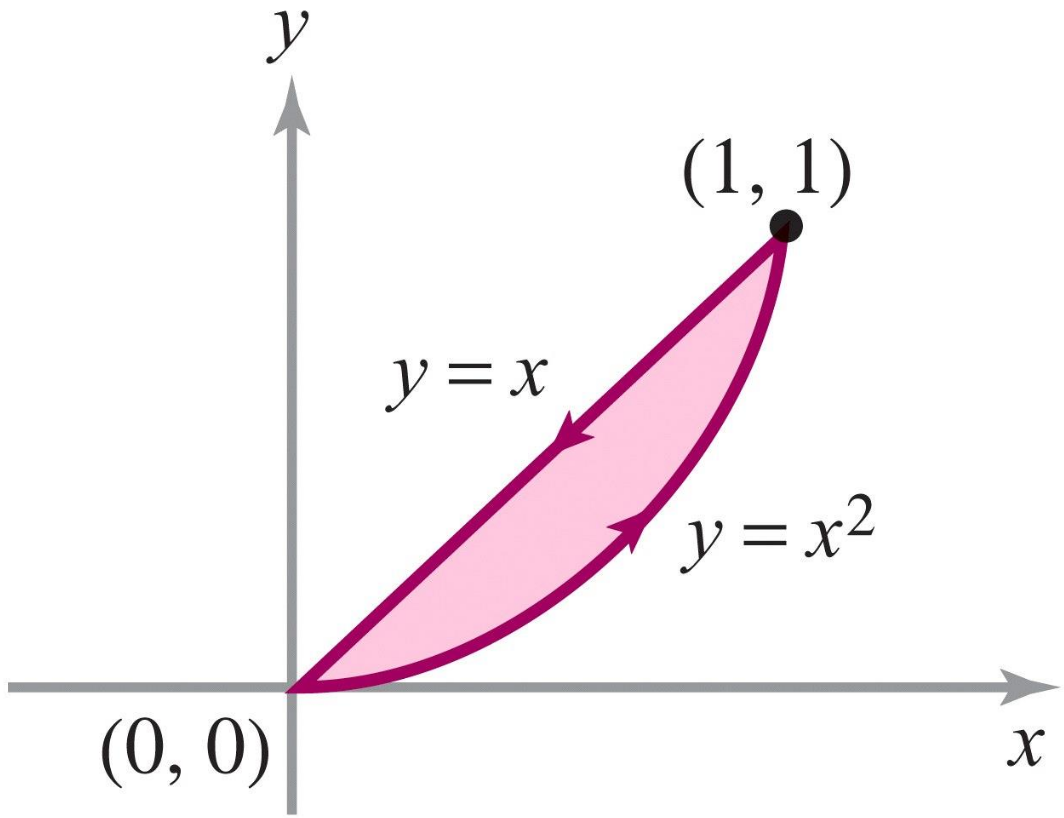
$$\int_{\partial D} \mathbf{F} \cdot d\mathbf{s} = \iint_D (\operatorname{curl} \mathbf{F}) \cdot \mathbf{k}\, dA = \iint_D (\nabla \times \mathbf{F}) \cdot \mathbf{k}\, dA$$

(see Figure 8.1.7).



$$(\nabla \times \mathbf{F}) \cdot \mathbf{k} = \partial Q/\partial x - \partial P/\partial y$$

Let $\mathbf{F} = (xy^2,\ y + x)$. Integrate $(\nabla \times \mathbf{F}) \cdot \mathbf{k}$ over the region in the first quadrant bounded by the curves $y = x^2$ and $y = x$.



Let $\mathbf{F} = (xy^2, y + x)$. Integrate $(\nabla \times \mathbf{F}) \cdot \mathbf{k}$ over the region in the first quadrant bounded by the curves $y = x^2$ and $y = x$.

Method 1. We first compute the curl

$$\nabla \times \mathbf{F} = \left(0, 0, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) = (1 - 2xy)\mathbf{k}.$$

Thus, $(\nabla \times \mathbf{F}) \cdot \mathbf{k} = 1 - 2xy$. This can be integrated over the given region D (see Figure 8.1.8) using an iterated integral as follows:

$$\begin{aligned}\iint_D (\nabla \times \mathbf{F}) \cdot \mathbf{k}\, dx\, dy &= \int_0^1 \int_{x^2}^{x} (1 - 2xy)\, dy\, dx = \int_0^1 [y - xy^2]\big|_{x^2}^{x}\, dx \\ &= \int_0^1 [x - x^3 - x^2 + x^5]\, dx = \frac{1}{2} - \frac{1}{4} - \frac{1}{3} + \frac{1}{6} = \frac{1}{12}.\end{aligned}$$

Let $\mathbf{F} = (xy^2, y + x)$. Integrate $(\nabla \times \mathbf{F}) \cdot \mathbf{k}$ over the region in the first quadrant bounded by the curves $y = x^2$ and $y = x$.

Method 2. Here we use Theorem 3 to obtain

$$\iint_D (\nabla \times \mathbf{F}) \cdot \mathbf{k}\, dx\, dy = \int_{\partial D} \mathbf{F} \cdot d\mathbf{s}.$$

The line integral of $\mathbf{F}$ along the curve $y = x$ from left to right is

$$\int_0^1 F_1\, dx + F_2\, dy = \int_0^1 (x^3 + 2x)\ dx = \frac{1}{4} + 1 = \frac{5}{4}.$$

Along the curve $y = x^2$ we get

$$\int_0^1 F_1\, dx + F_2\, dy = \int_0^1 x^5\, dx + (x + x^2)(2x\, dx) = \frac{1}{6} + \frac{2}{3} + \frac{1}{2} = \frac{4}{3}.$$

Thus, remembering that the integral along $y = x$ is to be taken from right to left, as in Figure 8.1.8,

$$\int_{\partial D} \mathbf{F} \cdot d\mathbf{s} = \frac{4}{3} - \frac{5}{4} = \frac{1}{12}. \qquad \blacktriangle$$

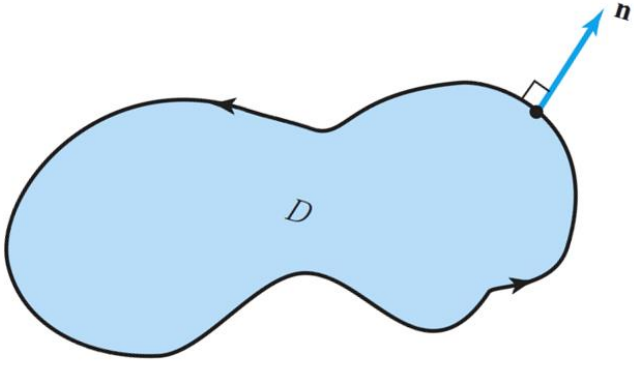
Teorema de Green utilitzant la divergència

THEOREM 4: Divergence Theorem in the Plane Let $D \subset \mathbb{R}^2$ be a region to which Green's theorem applies and let ∂D be its boundary. Let $\mathbf{n}$ denote the outward unit normal to ∂D. If $\mathbf{c}\colon [a, b] \to \mathbb{R}^2$, $t \mapsto \mathbf{c}(t) = (x(t), y(t))$ is a positively oriented parametrization of ∂D, $\mathbf{n}$ is given by

$$\mathbf{n} = \frac{(y'(t), -x'(t))}{\sqrt{[x'(t)]^2 + [y'(t)]^2}}$$

(see Figure 8.1.9). Let $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ be a C^1 vector field on D. Then

$$\int_{\partial D} \mathbf{F} \cdot \mathbf{n}\, ds = \iint_D \operatorname{div} \mathbf{F}\, dA .$$



proof Recall that $\mathbf{c}'(t) = (x'(t), y'(t))$ is tangent to ∂D, and note that $\mathbf{n} \cdot \mathbf{c}' = 0$. Thus, $\mathbf{n}$ is normal to the boundary. The sign of $\mathbf{n}$ is chosen to make it correspond to the *outward* (rather than the inward) direction. By the definition of the line integral (see Section 7.2),

$$\begin{aligned}
\int_{\partial D} \mathbf{F} \cdot \mathbf{n}\, ds &= \int_a^b \frac{P(x(t), y(t))\, y'(t) - Q(x(t), y(t))\, x'(t)}{\sqrt{[x'(t)]^2 + [y'(t)]^2}} \sqrt{[x'(t)]^2 + [y'(t)]^2}\; dt \\
&= \int_a^b [P(x(t), y(t))\, y'(t) - Q(x(t), y(t))\, x'(t)]\; dt \\
&= \int_{\partial D} P\,dy - Q\,dx.
\end{aligned}$$

By Green's theorem, this equals

$$\iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dx\, dy = \iint_D \operatorname{div} \mathbf{F}\, dA. \qquad \blacksquare$$

Let $\mathbf{F} = y^3\mathbf{i} + x^5\mathbf{j}$. Compute the integral of the normal component of $\mathbf{F}$ around the unit square.

This can be done using the divergence theorem. Indeed

$$\int_{\partial D} \mathbf{F} \cdot \mathbf{n}\, ds = \iint_D \operatorname{div} \mathbf{F}\, dA.$$

But $\operatorname{div} \mathbf{F} = 0$, and so the integral is zero. ▲

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Chapter 8:
The Integral Theorems of Vector Analysis

8.2 Stokes' Theorem

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Key Points in this Section.

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1. **Statement of Stoke's Theorem.** Let S be the oriented surface defined by the graph of a C^2 function $z = f(x, y)$, where $(x, y) \in D$, a region in the plane to which Green's theorem applies, and let F be a C^1 vector field on a region containing the surface. If ∂S denotes the oriented boundary curve of S, then

$$\iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{s}.$$

2. The main idea in the proof of this result is to reduce the problem to Green's theorem over the region D by everywhere substituting z in terms of x and y.

3. The same statement holds for parametrized surfaces as well, and the main idea of the proof is the same; this time one reduces it to Green's theorem by substituting for x, y and z their expressions in terms of the surface parameters, u and v.

4. The default orientation for graphs is that the surface is oriented by the upward pointing normal vector; that is, by

$$\mathbf{n} = -\frac{\partial g}{\partial y}\mathbf{i} - \frac{\partial g}{\partial x}\mathbf{j} + \mathbf{k}$$

(note that this vector need not be a *unit vector*). One traverses the boundary in the same way as one traverses the boundary in the domain D as in Green's theorem.

5. For a parametrized surface, if one's head is pointing in the direction of the chosen normal vector (which determines the orientation of the surface), and if one walks along the boundary curve ∂S in the correct oriented direction, then the surface is on your left. (If the surface is on your right, then you are going in the wrong direction and you must change direction or change the orientation of the surface).

6. Stokes' Theorem together with the mean value theorem gives the interpretation of the curl of a vector field $\mathbf{F}$ as the circulation per unit area. That is, if we choose a point P and a *unit* vector $\mathbf{n}$ at this point, then

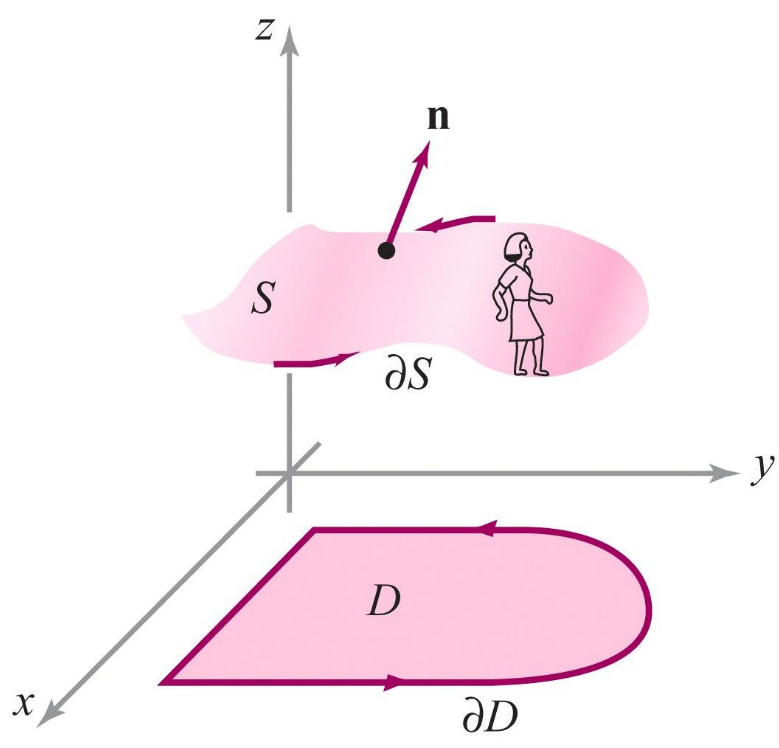
$$(\operatorname{curl}\mathbf{F}(\mathrm{P}))\cdot\mathbf{n} = \lim_{\rho\to 0}\frac{1}{A(S_\rho)}\int_{\partial S_\rho}\mathbf{F}\cdot d\mathbf{s}$$

where S_ρ is a disk of radius ρ in the plane perpendicular to $\mathbf{n}$ and centered at the point P and $A(S_\rho) = \pi\rho^2$ is its area (shapes other than disks can be used just as well).

7. The interpretation of the curl as circulation per unit area is useful in deriving formulas for the curl in cylindrical and spherical coordinates.

THEOREM 5: Stokes' Theorem for Graphs Let S be the oriented surface defined by a C^2 function $z = f(x, y)$, where $(x, y) \in D$, a region to which Green's theorem applies, and let $\mathbf{F}$ be a C^1 vector field on S. Then if ∂S denotes the oriented boundary curve of S as just defined, we have

$$\iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{s}.$$



proof If $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$, then

$$\operatorname{curl} \mathbf{F} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right)\mathbf{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right)\mathbf{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)\mathbf{k}.$$

We know

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \left[F_1\left(-\frac{\partial z}{\partial x}\right) + F_2\left(-\frac{\partial z}{\partial y}\right) + F_3\right] dx\, dy, \qquad (1)$$

Therefore, we use formula (1) to write

$$\iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S} = \iint_D \left[\left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right)\left(-\frac{\partial z}{\partial x}\right) + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right)\left(-\frac{\partial z}{\partial y}\right) + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)\right] dA. \qquad (2)$$

On the other hand,

$$\int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{p}} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{p}} F_1\, dx + F_2\, dy + F_3\, dz,$$

where $\mathbf{p}\colon [a, b] \to \mathbb{R}^3$, $\mathbf{p}(t) = (x(t), y(t), f(x(t), y(t)))$ is the orientation-preserving parametrization of the oriented simple closed curve ∂S discussed earlier. Thus,

$$\int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = \int_a^b \left(F_1\frac{dx}{dt} + F_2\frac{dy}{dt} + F_3\frac{dz}{dt}\right) dt. \qquad (3)$$

By the chain rule,

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}.$$

Substituting this expression into equation (3), we obtain

$$\begin{aligned}
\int_{\partial S} \mathbf{F} \cdot d\mathbf{s} &= \int_a^b \left[\left(F_1 + F_3 \frac{\partial z}{\partial x}\right)\frac{dx}{dt} + \left(F_2 + F_3 \frac{\partial z}{\partial y}\right)\frac{dy}{dt}\right] dt \\
&= \int_{\mathbf{c}} \left(F_1 + F_3 \frac{\partial z}{\partial x}\right) dx + \left(F_2 + F_3 \frac{\partial z}{\partial y}\right) dy \qquad (4) \\
&= \int_{\partial D} \left(F_1 + F_3 \frac{\partial z}{\partial x}\right) dx + \left(F_2 + F_3 \frac{\partial z}{\partial y}\right) dy.
\end{aligned}$$

Applying Green's theorem to equation (4) yields (we are assuming that Green's theorem applies to D)

$$\iint_D \left[\frac{\partial(F_2 + F_3\,\partial z/\partial y)}{\partial x} - \frac{\partial(F_1 + F_3\,\partial z/\partial x)}{\partial y}\right] dA.$$

Now we use the chain rule, remembering that F_1, F_2, and F_3 are functions of x, y, and z and that z is a function of x and y, to obtain

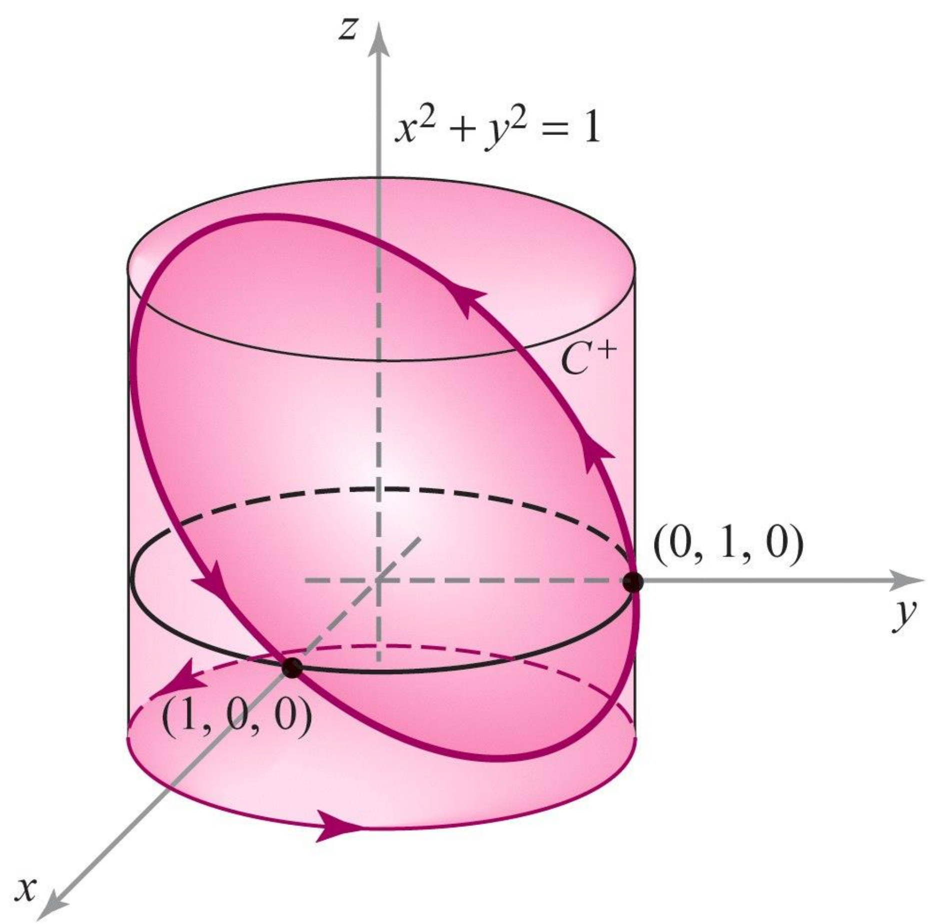
$$\iint_D \left[\left(\frac{\partial F_2}{\partial x} + \frac{\partial F_2}{\partial z}\frac{\partial z}{\partial x} + \frac{\partial F_3}{\partial x}\frac{\partial z}{\partial y} + \frac{\partial F_3}{\partial z}\frac{\partial z}{\partial x}\frac{\partial z}{\partial y} + F_3 \frac{\partial^2 z}{\partial x \partial y} \right) \right.$$
$$\left. - \left(\frac{\partial F_1}{\partial y} + \frac{\partial F_1}{\partial z}\frac{\partial z}{\partial y} + \frac{\partial F_3}{\partial y}\frac{\partial z}{\partial x} + \frac{\partial F_3}{\partial z}\frac{\partial z}{\partial y}\frac{\partial z}{\partial x} + F_3 \frac{\partial^2 z}{\partial y \partial x} \right) \right] dA.$$

Because mixed partials are equal, the last two terms in each parenthesis cancel each other, and we can rearrange terms to obtain the integral of equation (2), which completes the proof. ■

Use Stokes' theorem to evaluate the line integral

$$\int_C -y^3\,dx + x^3\,dy - z^3\,dz,$$

where C is the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $x + y + z = 1$, and the orientation on C corresponds to counterclockwise motion in the xy plane.



The curve C bounds the surface S defined by the equation $z = 1 - x - y = f(x, y)$ for (x, y) in the set $D = \{(x, y) \mid x^2 + y^2 \leq 1\}$ (Figure 8.2.2). We set $\mathbf{F} = -y^3\mathbf{i} + x^3\mathbf{j} - z^3\mathbf{k}$, which has curl $\nabla \times \mathbf{F} = (3x^2 + 3y^2)\mathbf{k}$. Then, by Stokes' theorem, the line integral is equal to the surface integral

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}.$$

But $\nabla \times \mathbf{F}$ has only a $\mathbf{k}$ component. Thus, by formula (1) we have

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \iint_D (3x^2 + 3y^2)\ dx\,dy.$$

This integral can be evaluated by changing to polar coordinates. Doing this, we get

$$3\iint_D (x^2 + y^2)\ dx\,dy = 3\int_0^1 \int_0^{2\pi} r^2 \cdot r\,d\theta\,dr = 6\pi \int_0^1 r^3\,dr = \frac{6\pi}{4} = \frac{3\pi}{2}.$$

Let us verify this result by *directly* evaluating the line integral

$$\int_C -y^3\,dx + x^3\,dy - z^3\,dz.$$

We can parametrize the curve ∂D by the equations

$$x = \cos t, \qquad y = \sin t, \qquad z = 0, \qquad 0 \le t \le 2\pi.$$

The curve C is therefore parametrized by the equations

$$x = \cos t, \qquad y = \sin t, \qquad z = 1 - \sin t - \cos t, \qquad 0 \le t \le 2\pi.$$

Thus,

$$\int_C -y^3\,dx + x^3\,dy - z^3\,dz$$

$$= \int_0^{2\pi} (\cos^4 t + \sin^4 t)\;dt - \int_0^{2\pi} (1 - \sin t - \cos t)^3(-\cos t + \sin t)\;dt.$$

The second integrand is of the form $u^3\,du$, where $u = 1 - \sin t - \cos t$, and thus the integral is equal to

$$\frac{1}{4}[(1 - \sin t - \cos t)^4]_0^{2\pi} = 0.$$

Hence, we are left with

$$\int_0^{2\pi} (\cos^4 t + \sin^4 t)\ dt.$$

This integral can be evaluated using formulas (18) and (19) of the table of integrals. We can also proceed as follows. Using the trigonometric identities

$$\sin^2 t = \frac{1 - \cos 2t}{2}, \qquad \cos^2 t = \frac{1 + \cos 2t}{2},$$

substituting and squaring these expressions, we reduce the preceding integral to

$$\frac{1}{2}\int_0^{2\pi} (1 + \cos^2 2t)\ dt = \pi + \frac{1}{2}\int_0^{2\pi} \cos^2 2t\ dt.$$

Again using the identity $\cos^2 2t = (1 + \cos 4t)/2$, we find

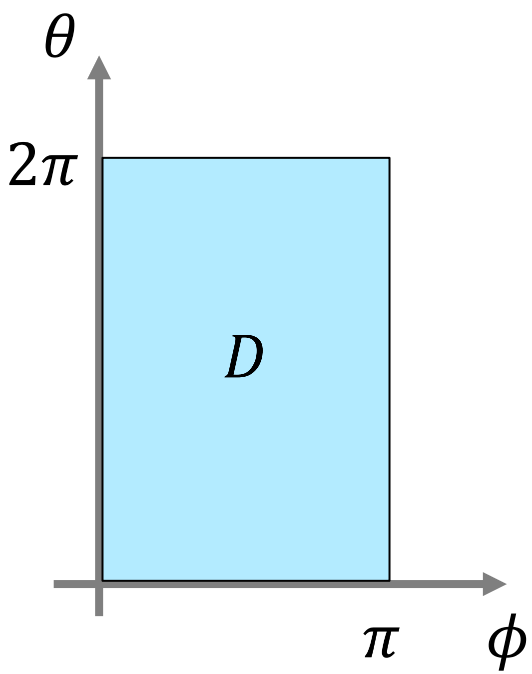
$$\begin{aligned}\pi + \frac{1}{4}\int_0^{2\pi} (1 + \cos 4t)\ dt &= \pi + \frac{1}{4}\int_0^{2\pi} dt + \frac{1}{4}\int_0^{2\pi} \cos 4t\ dt \\ &= \pi + \frac{\pi}{2} + 0 = \frac{3\pi}{2}.\end{aligned}$$

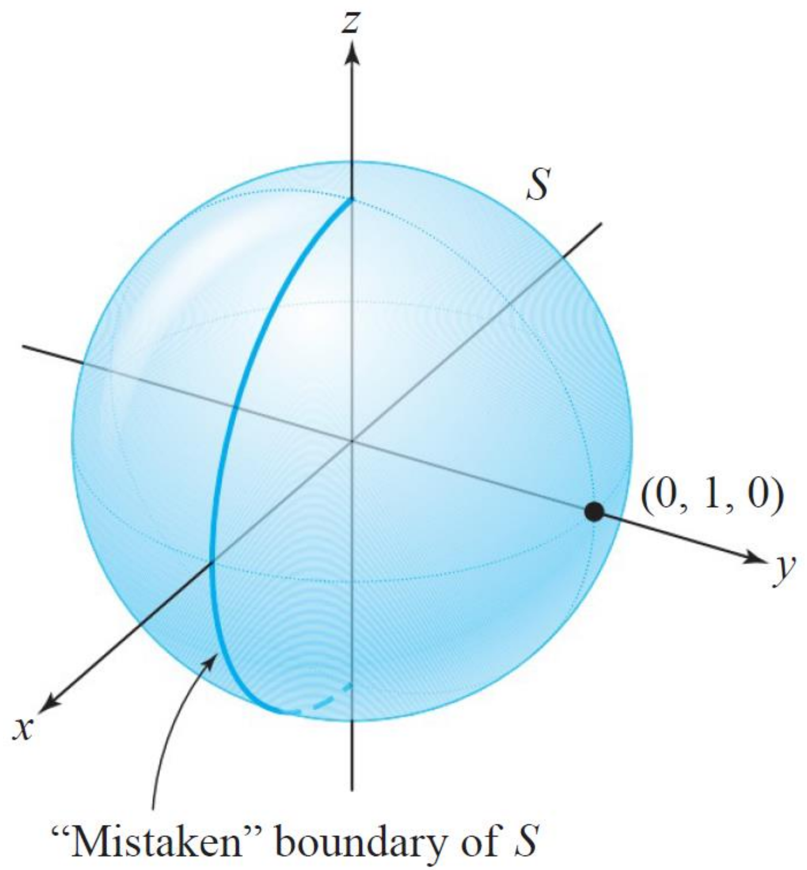
▲

Stokes' Theorem for Parametrized Surfaces

Suppose $\mathbf{\Phi}\colon D \to \mathbb{R}^3$ is a parametrization of a surface S and $\mathbf{c}(t) = (u(t), v(t))$ is a parametrization of ∂D. We might be tempted to define ∂S as the curve parametrized by $t \mapsto \mathbf{p}(t) = \mathbf{\Phi}(u(t), v(t))$. However, with this definition, ∂S might not be the boundary of S in any reasonable geometric sense.

We can get around this difficulty by assuming that $\mathbf{\Phi}$ is one-to-one on all of D.



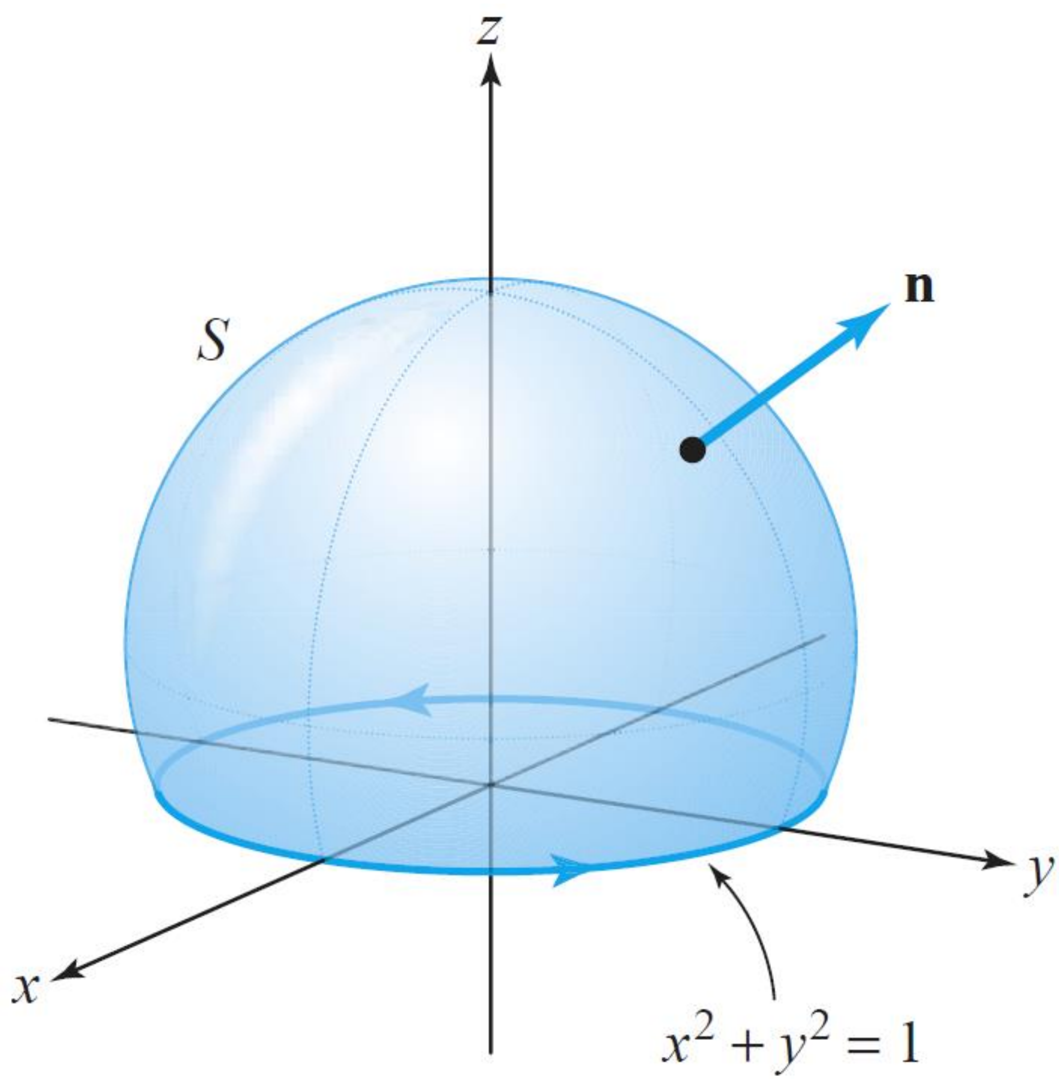


THEOREM 6: Stokes' Theorem: Parametrized Surfaces Let S be an oriented surface defined by a one-to-one parametrization $\mathbf{\Phi}\colon D \subset \mathbb{R}^2 \to S$, where D is a region to which Green's theorem applies. Let ∂S denote the oriented boundary of S and let $\mathbf{F}$ be a C^1 vector field on S. Then

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{s}.$$

If S has no boundary, and this includes surfaces such as the sphere, then the integral on the left is zero (see Exercise 17).

Let S be the surface shown in Figure 8.2.4, with the indicated orientation. Let $\mathbf{F} = y\mathbf{i} - x\mathbf{j} + e^{xz}\mathbf{k}$. Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$.



Let S be the surface shown in Figure 8.2.4, with the indicated orientation. Let $\mathbf{F} = y\mathbf{i} - x\mathbf{j} + e^{xz}\mathbf{k}$. Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$.

This surface could be parametrized using spherical coordinates based at the center of the sphere. However, we need not explicitly find $\mathbf{\Phi}$ in order to solve this problem. By Theorem 6, $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{s}$, and so if we parametrize ∂S by $x(t) = \cos t$, $y(t) = \sin t$, $0 \le t \le 2\pi$, we determine

$$\int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = \int_0^{2\pi} \left(y\frac{dx}{dt} - x\frac{dy}{dt} \right) dt = \int_0^{2\pi} (-\sin^2 t - \cos^2 t)\ dt = -\int_0^{2\pi} dt = -2\pi$$

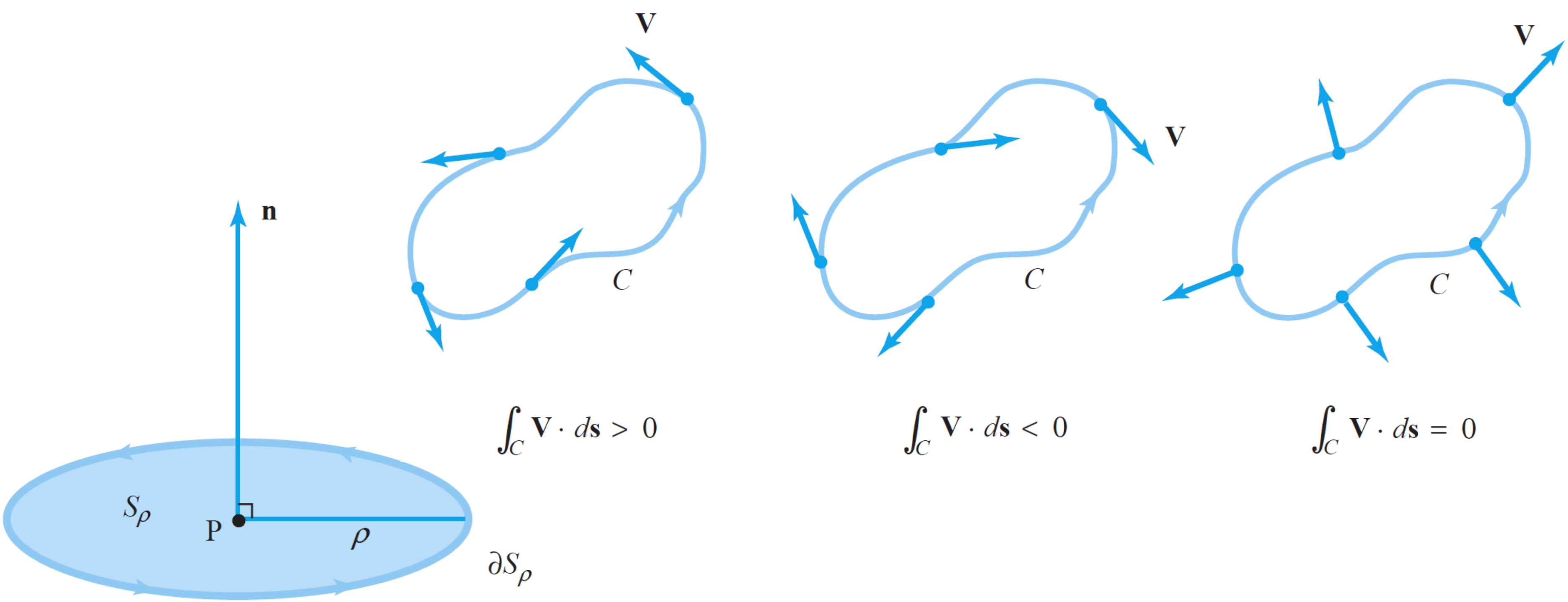
and therefore $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = -2\pi$. ▲

Physical interpretation of the Curl: Circulation

Suppose $\mathbf{V}$ represents the velocity vector field of a fluid. Consider a point P and a unit vector $\mathbf{n}$. Let S_ρ denote the disc of radius ρ and center P, which is perpendicular to $\mathbf{n}$. By Stokes' theorem,

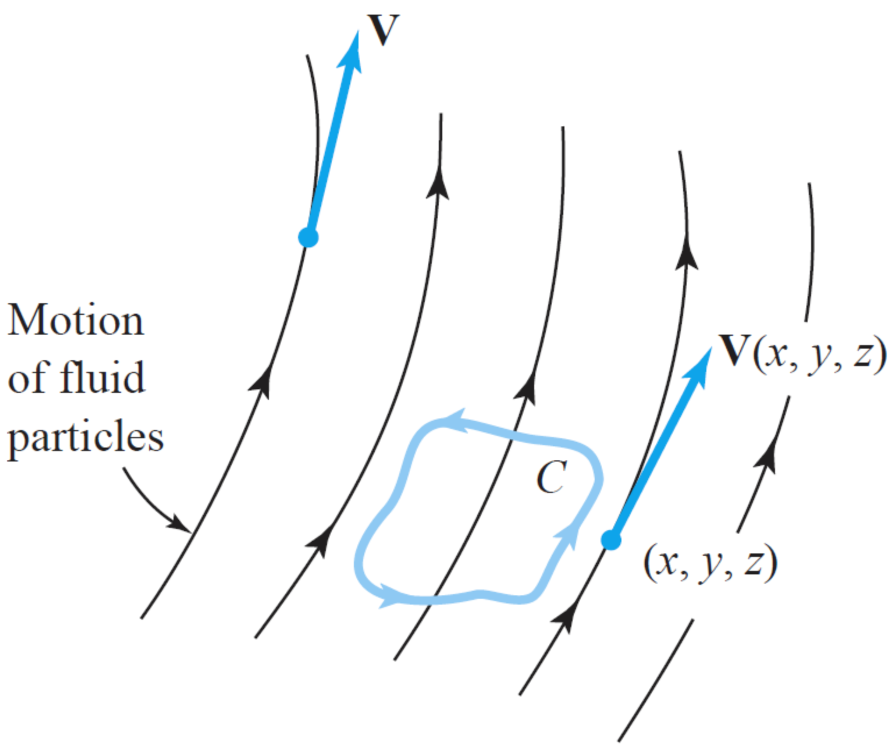
$$\iint_{S_\rho} \operatorname{curl} \mathbf{V} \cdot d\mathbf{S} = \iint_{S_\rho} \operatorname{curl} \mathbf{V} \cdot \mathbf{n}\; dS = \int_{\partial S_\rho} \mathbf{V} \cdot d\mathbf{s},$$

where ∂S_ρ has the orientation induced by $\mathbf{n}$ (see Figure 8.2.5).

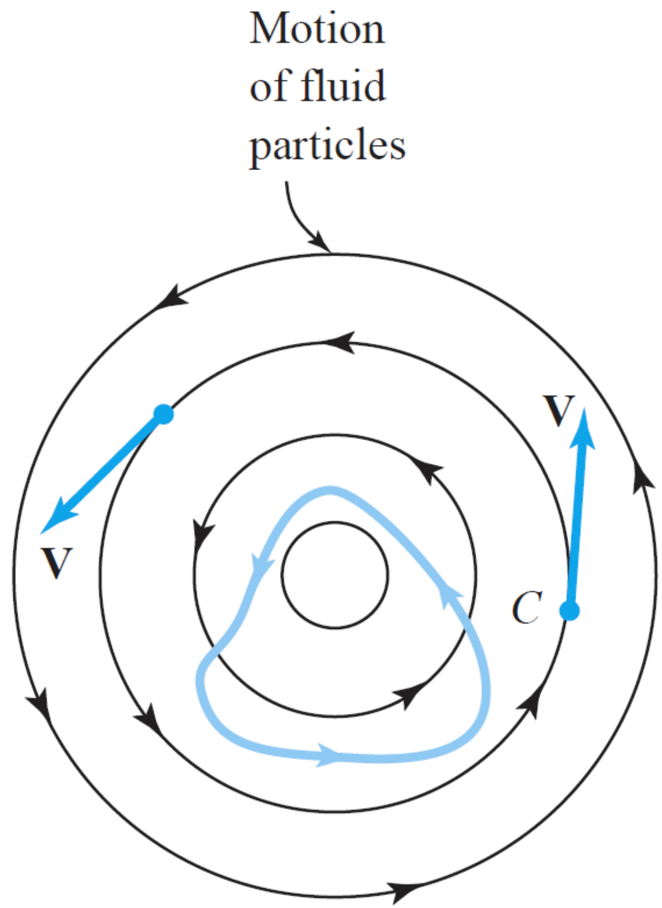


Using the mean-value theorem of integrals

$$\text{curl } \mathbf{V}(\mathrm{P}) \cdot \mathbf{n} = \lim_{\rho \to 0} \frac{1}{A(S_\rho)} \int_{\partial S_\rho} \mathbf{V} \cdot d\mathbf{s}$$



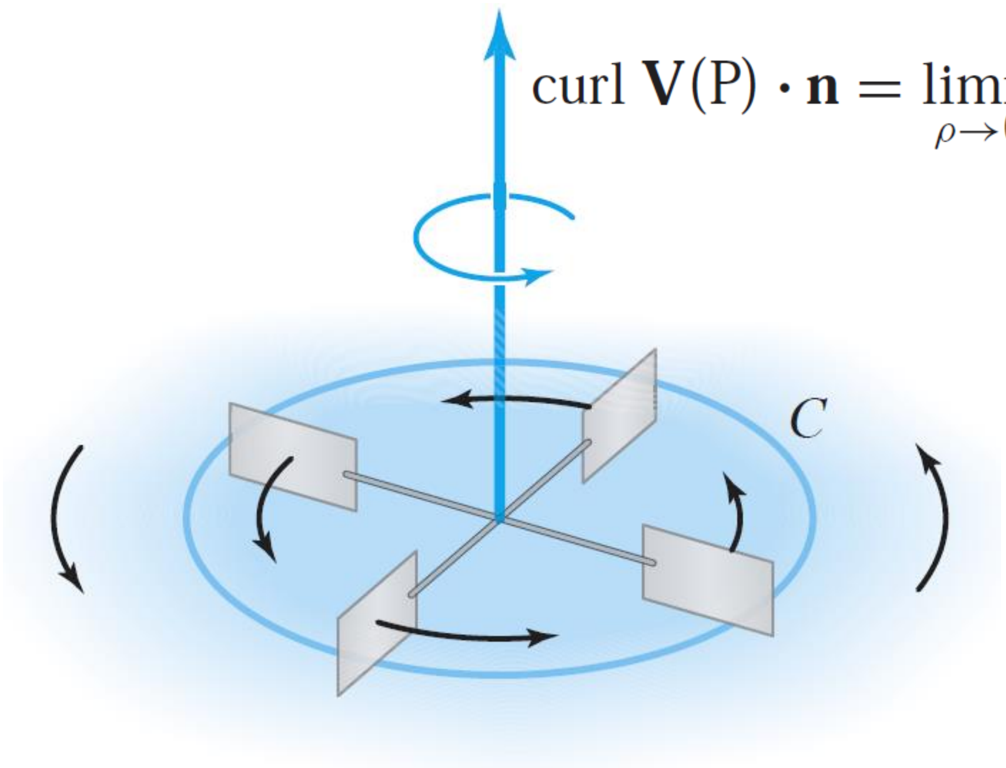
Zero circulation about C



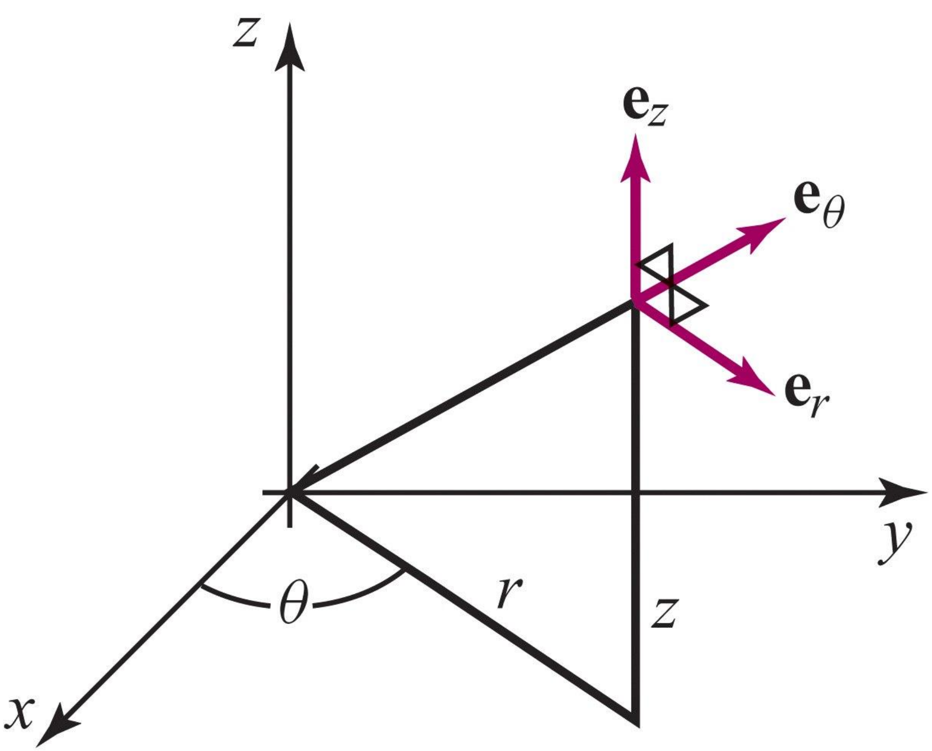
Nonzero circulation about C
("whirlpool")

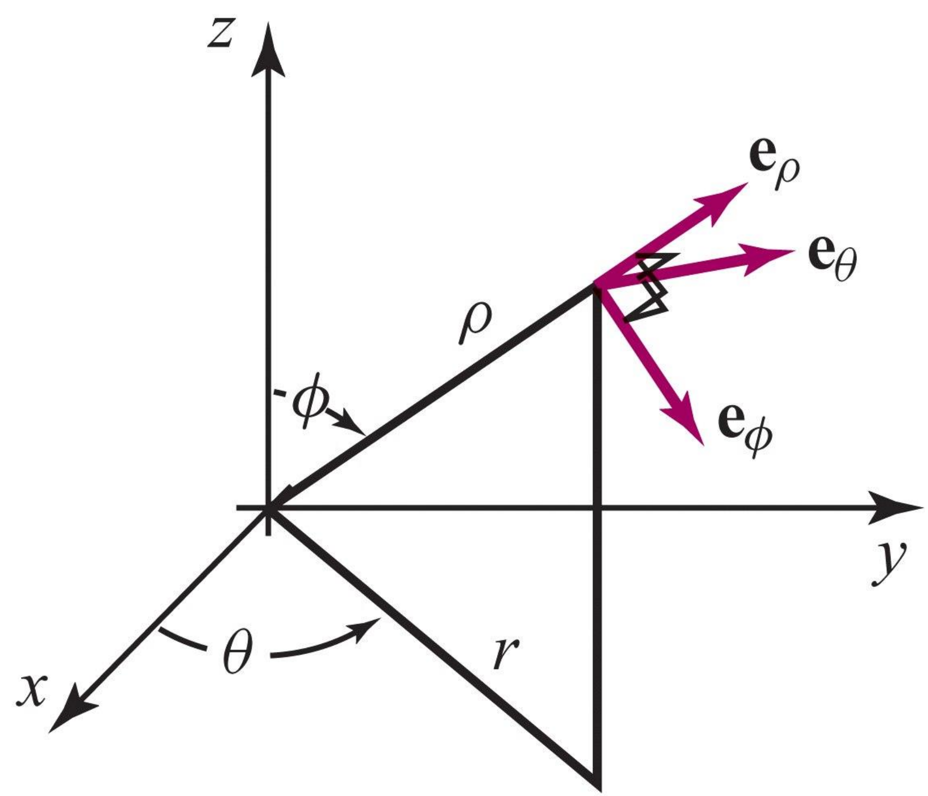
Circulation and Curl The dot product of curl $\mathbf{V}(\mathrm{P})$ with a unit vector $\mathbf{n}$, namely, curl $\mathbf{V}(\mathrm{P}) \cdot \mathbf{n}$, equals the circulation of $\mathbf{V}$ per unit area at P on a surface perpendicular to $\mathbf{n}$.

$$\operatorname{curl} \mathbf{V}(\mathrm{P}) \cdot \mathbf{n} = \operatorname*{limit}_{\rho \to 0} \frac{1}{A(S_\rho)} \int_{\partial S_\rho} \mathbf{V} \cdot d\mathbf{s}$$



Orthonormal vectors in cylindrical and spherical coordinates





Cylindrical coordinates

$$\mathbf{F} = F_r \mathbf{e}_r + F_\theta \mathbf{e}_\theta + F_z \mathbf{e}_z$$

$$\nabla f = \frac{\partial f}{\partial r}\mathbf{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\mathbf{e}_\theta + \frac{\partial f}{\partial z}\mathbf{e}_z$$

$$\nabla \cdot \mathbf{F} = \frac{1}{r}\left[\frac{\partial}{\partial r}(rF_r) + \frac{\partial F_\theta}{\partial \theta} + \frac{\partial}{\partial z}(rF_z)\right]$$

$$\nabla \times \mathbf{F} = \frac{1}{r}\begin{vmatrix} \mathbf{e}_r & r\mathbf{e}_\theta & \mathbf{e}_z \\ \dfrac{\partial}{\partial r} & \dfrac{\partial}{\partial \theta} & \dfrac{\partial}{\partial z} \\ F_r & rF_\theta & F_z \end{vmatrix}$$

Spherical coordinates

$$\mathbf{F} = F_\rho \mathbf{e}_\rho + F_\phi \mathbf{e}_\phi + F_\theta \mathbf{e}_\theta$$

$$\nabla f = \frac{\partial f}{\partial \rho}\,\mathbf{e}_\rho + \frac{1}{\rho}\frac{\partial f}{\partial \phi}\,\mathbf{e}_\phi + \frac{1}{\rho \sin\phi}\frac{\partial f}{\partial \theta}\,\mathbf{e}_\theta$$

$$\nabla \cdot \mathbf{F} = \frac{1}{\rho^2}\frac{\partial}{\partial \rho}(\rho^2 F_\rho) + \frac{1}{\rho \sin\phi}\frac{\partial}{\partial \phi}(\sin\phi\, F_\phi) + \frac{1}{\rho \sin\phi}\frac{\partial F_\theta}{\partial \theta}$$

$$\nabla \times \mathbf{F} = \left[\frac{1}{\rho \sin\phi}\frac{\partial}{\partial \phi}(\sin\phi\, F_\theta) - \frac{1}{\rho \sin\phi}\frac{\partial F_\phi}{\partial \theta}\right]\mathbf{e}_\rho + \left[\frac{1}{\rho \sin\phi}\frac{\partial F_\rho}{\partial \theta} - \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho F_\theta)\right]\mathbf{e}_\phi + \left[\frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho F_\phi) - \frac{1}{\rho}\frac{\partial F_\rho}{\partial \phi}\right]\mathbf{e}_\theta$$

Faraday's law from Maxwell equations

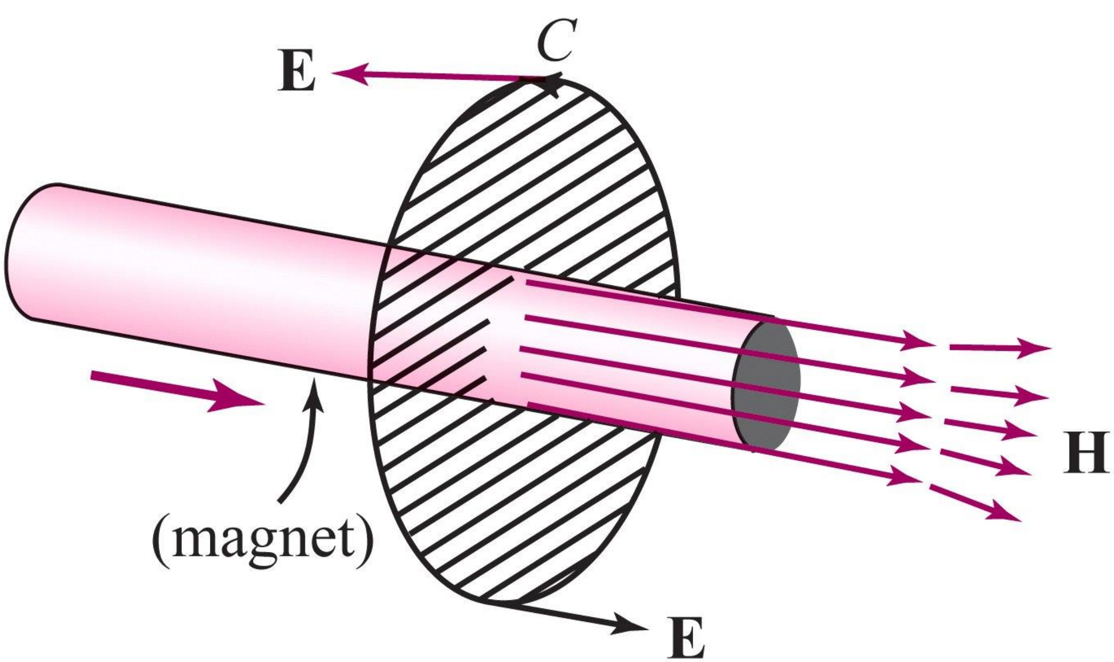
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{H}}{\partial t}$$

$$\int_C \mathbf{E} \cdot d\mathbf{s} = \text{voltage around } C,$$

$$\iint_S \mathbf{H} \cdot d\mathbf{S} = \text{magnetic flux across } S.$$

By Stokes' theorem,

$$\int_C \mathbf{E} \cdot d\mathbf{s} = \iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S}$$



Faraday's law from Maxwell equations

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{H}}{\partial t}$$

$$\int_C \mathbf{E} \cdot d\mathbf{s} = \text{voltage around } C,$$

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By Stokes' theorem,

$$\int_C \mathbf{E} \cdot d\mathbf{s} = \iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S}$$

Assuming that we can move $\partial/\partial t$ under the integral sign, we get

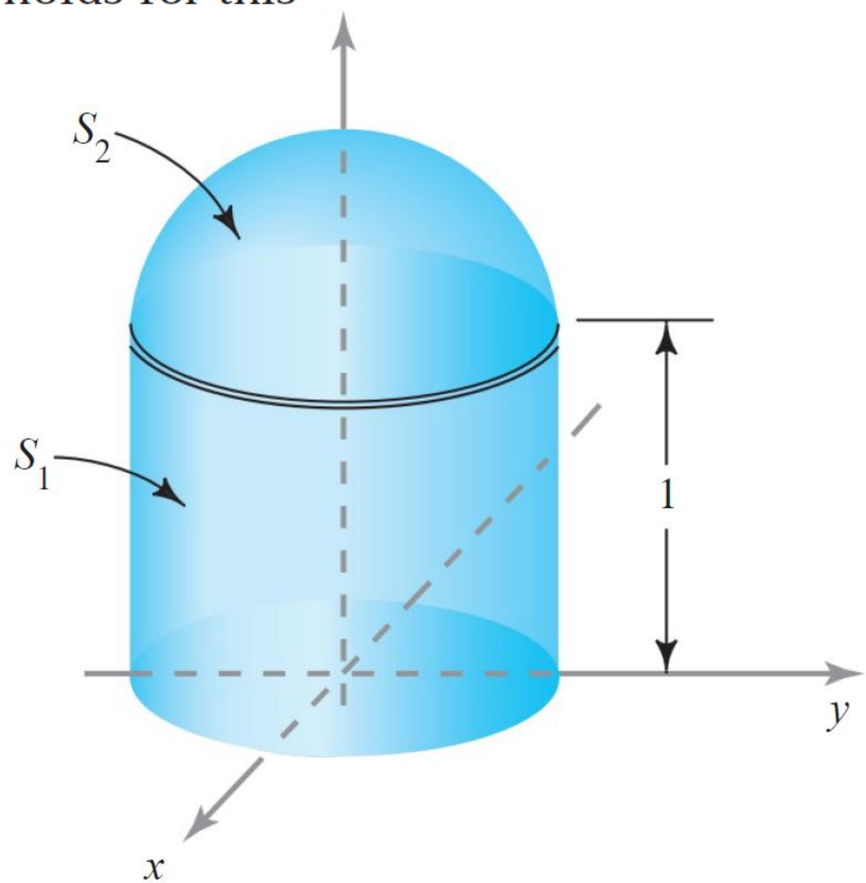
$$-\frac{\partial}{\partial t} \iint_S \mathbf{H} \cdot d\mathbf{S} = \iint_S -\frac{\partial \mathbf{H}}{\partial t} \cdot d\mathbf{S} = \iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = \int_C \mathbf{E} \cdot d\mathbf{s}$$

and so

$$\int_C \mathbf{E} \cdot d\mathbf{s} = -\frac{\partial}{\partial t} \iint_S \mathbf{H} \cdot d\mathbf{S},$$

which is Faraday's law. ▲

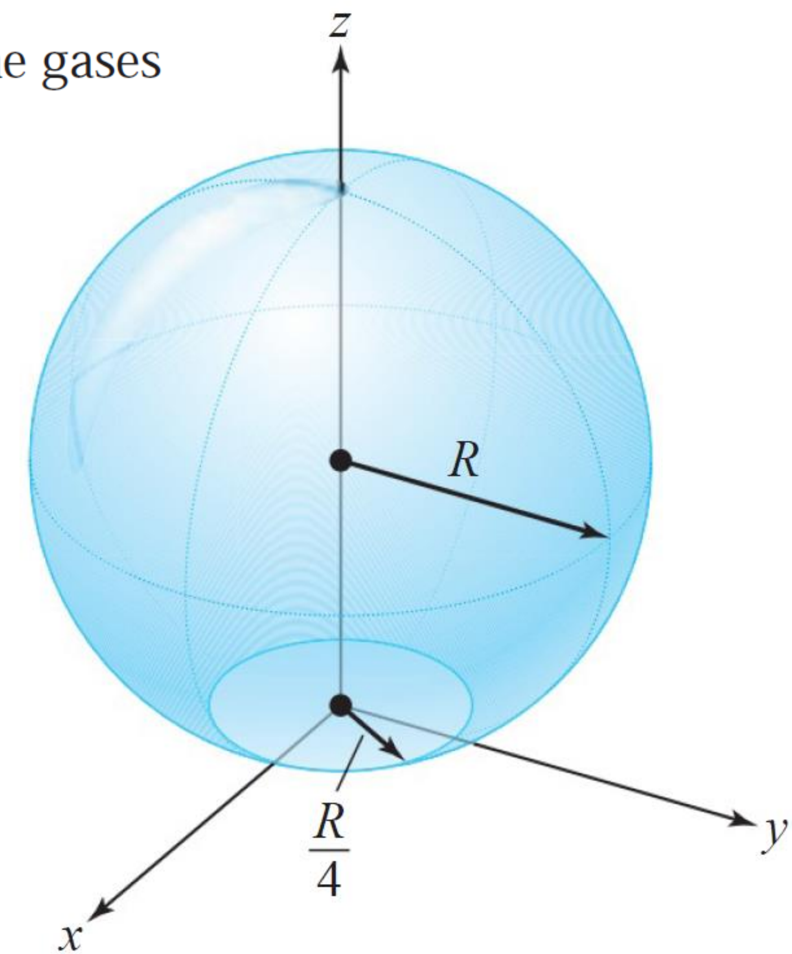
Let S be the capped cylindrical surface shown in Figure 8.2.13. S is the union of two surfaces, S_1 and S_2, where S_1 is the set of (x, y, z) with $x^2 + y^2 = 1$, $0 \leq z \leq 1$, and S_2 is the set of (x, y, z) with $x^2 + y^2 + (z-1)^2 = 1$, $z \geq 1$. Set $\mathbf{F}(x, y, z) = (zx + z^2 y + x)\mathbf{i} + (z^3 yx + y)\mathbf{j} + z^4 x^2 \mathbf{k}$. Compute $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$. (HINT: Stokes' theorem holds for this surface.)



A hot-air balloon has the truncated spherical shape shown in Figure 8.2.14. The hot gases escape through the porous envelope with a velocity vector field

$$\mathbf{V}(x, y, z) = \nabla \times \mathbf{\Phi}(x, y, z) \quad \text{where} \quad \mathbf{\Phi}(x, y, z) = -y\mathbf{i} + x\mathbf{j}.$$

If $R = 5$, compute the volume flow rate of the gases through the surface.



Jerrold E. Marsden and Anthony J. Tromba

Vector Calculus

Fifth Edition

Chapter 8:
The Integral Theorems of Vector Analysis

8.3 Conservative Fields

8.3 Conservative Fields

Key Points in this Section.

1. The main result in this section states that the following statements concerning a vector field $\mathbf{F}$ defined and C^1 on all of $\mathbb{R}^3$ are equivalent:

   (a) The integral of $\mathbf{F}$ around any closed loop is zero

   (b) The integral of $\mathbf{F}$ from one point to another is independent of the path taken between those points

   (c) $\mathbf{F}$ is a gradient field

   (d) $\nabla \times \mathbf{F} = \mathbf{0}$.

2. A similar result holds in the plane (where the curl is interpreted as the scalar curl)

3. Stokes' theorem is used to show that if $\mathbf{F}$ is curl free, then its integral around a closed loop is zero.

4. If $\mathbf{F}$ is not defined at a finite number of points in $\mathbb{R}^3$, then the same result is true. This does not necessarily hold in the plane. (A counter example is given in Exercise 12).

5. Special Case: In the plane, a vector field $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ defined and C^1 everywhere, is a gradient if and only if

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

THEOREM 7: Conservative Fields Let $\mathbf{F}$ be a C^1 vector field defined on $\mathbb{R}^3$ except possibly for a finite number of points. The following conditions on $\mathbf{F}$ are all equivalent:

(i) For any oriented simple closed curve C, $\int_C \mathbf{F} \cdot d\mathbf{s} = 0$.

(ii) For any two oriented simple curves C_1 and C_2 that have the same endpoints,

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{s} = \int_{C_2} \mathbf{F} \cdot d\mathbf{s}.$$

(iii) $\mathbf{F}$ is the gradient of some function f; that is, $\mathbf{F} = \nabla f$ (and if $\mathbf{F}$ has one or more exceptional points where it fails to be defined, f is also undefined there).

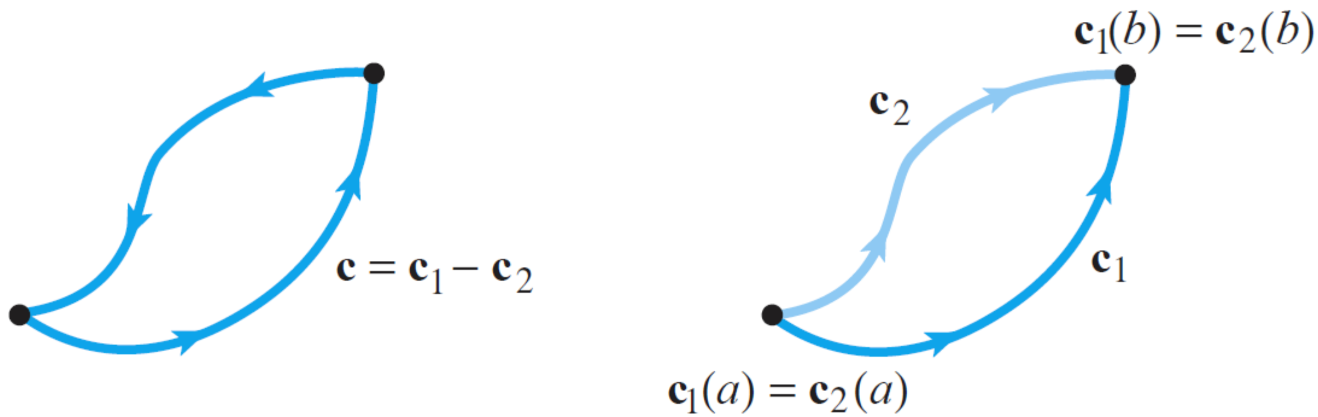
(iv) $\nabla \times \mathbf{F} = \mathbf{0}$.

A vector field satisfying one (and, hence, all) of the conditions (i)–(iv) is called a ***conservative vector field***.[6]

proof We shall establish the following chain of implications, which will prove the theorem:

$$(\text{i}) \Rightarrow (\text{ii}) \Rightarrow (\text{iii}) \Rightarrow (\text{iv}) \Rightarrow (\text{i}).$$

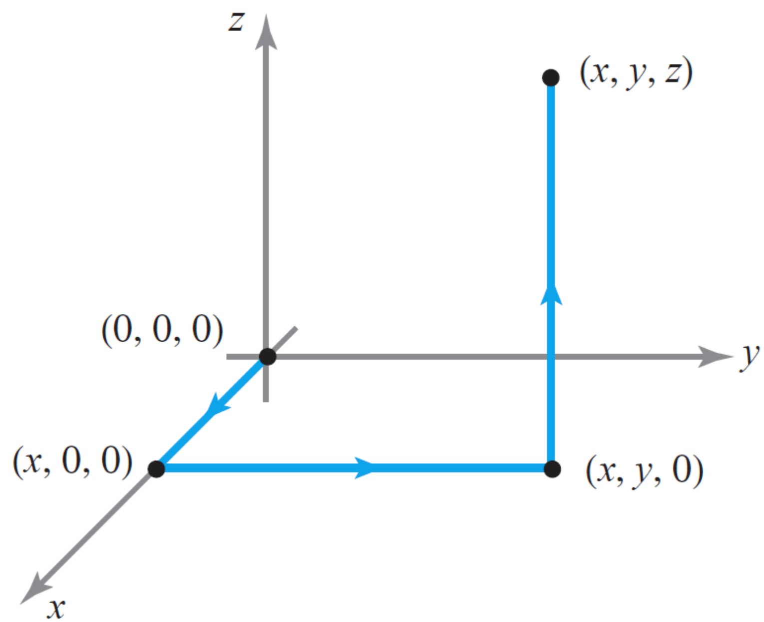
(i) $\Rightarrow$ (ii)



$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}_1} \mathbf{F} \cdot d\mathbf{s} - \int_{\mathbf{c}_2} \mathbf{F} \cdot d\mathbf{s} = 0$$

(ii) $\Rightarrow$ (iii)

$$\mathbf{F} = (F_1, F_2, F_3)$$



$$f(x, y, z) = \int_0^x F_1(t, 0, 0)\, dt + \int_0^y F_2(x, t, 0)\, dt + \int_0^z F_3(x, y, t)\, dt \quad \Rightarrow \frac{\partial f}{\partial z} = F_3$$

Using other paths from $(0,0,0)$ to (x, y, z) and the independence of the line integral from the path from (ii), we also get

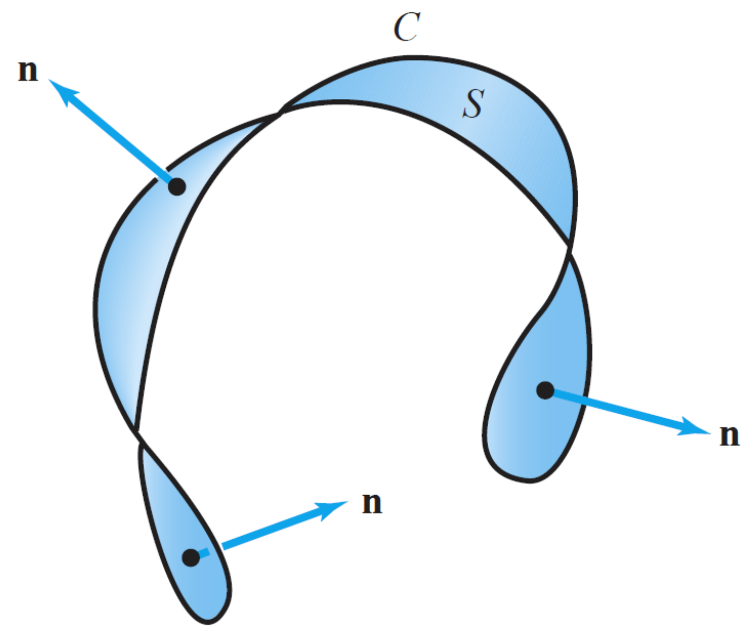
$$\frac{\partial f}{\partial y} = F_2, \qquad \frac{\partial f}{\partial x} = F_1$$

Thus $\boldsymbol{F} = \nabla f$

(iii) $\Rightarrow$ (iv)

$$\nabla \times \nabla f = \mathbf{0}$$

(iv) $\Rightarrow$ (i)



For a closed curve C

$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \; dS$$

which is zero since $\nabla \times \boldsymbol{F} = 0$

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Vector Calculus

Fifth Edition

Chapter 8:
The Integral Theorems of Vector Analysis

8.4 Gauss' Theorem

8.4 Gauss' Theorem

Key Points in this Section.

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1. If S is a closed surface enclosing a region W, we adopted the convention that $S = \partial W$ is given the ***outward orientation***, with outward unit normal denoted by $\mathbf{n}(x, y, z)$ at each point (x, y, z) of S. If we denote the surface with the opposite (inward) orientation by ∂W_{op}, then the associated unit normal direction for this orientation is $-\mathbf{n}$. Thus,

$$\iint_{\partial W} \mathbf{F} \cdot d\mathbf{S} = \iint_S (\mathbf{F} \cdot \mathbf{n}) dS = -\iint_S [\mathbf{F} \cdot (-\mathbf{n})] dS = -\iint_{\partial W_{\text{op}}} \mathbf{F} \cdot d\mathbf{S}.$$

2. ***Gauss' Divergence Theorem*** states that for a (symmetric, elementary) region W with boundary ∂W oriented by the outward pointing unit normal and if $\mathbf{F}$ is a smooth vector field defined on W, then

$$\iiint_W (\nabla \cdot \mathbf{F}) dV = \iint_{\partial W} \mathbf{F} \cdot d\mathbf{S}.$$

3. The ***key idea of the proof*** is to proceed in these steps:

   (a) Write $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ so that

   $$\nabla \cdot \mathbf{F} = \partial P/\partial x + \partial Q/\partial y + \partial R/\partial z.$$

   (b) Establish the separate identities

   $$\iiint_W \frac{\partial P}{\partial x} dV = \iint_{\partial W} P\mathbf{i} \cdot d\mathbf{S}$$
   $$\iiint_W \frac{\partial Q}{\partial y} dV = \iint_{\partial W} Q\mathbf{j} \cdot d\mathbf{S}$$
   $$\iiint_W \frac{\partial R}{\partial z} dV = \iint_{\partial W} R\mathbf{k} \cdot d\mathbf{S},$$

   which is parallel to what was done in the proof of Green's theorem.

   (c) Adding these identities gives the divergence theorem

   (d) To establish the above identities, proceed in a manner similar to Green's theorem, namely reduce the triple integral to a double + single integral and apply the fundamental theorem of calculus to the single integral.

(e) For the third identity (the one involving R), for instance, write the region as that between the graphs of two functions $z = f_2(x, y)$ and $z = f_1(x, y)$ over a region D in the xy-plane. Then,

$$\begin{aligned}\iiint_W \frac{\partial R}{\partial z} dV &= \iint_D \left[\int_{z=f_1(x,y)}^{z=f_2(x,y)} \frac{\partial R}{\partial z}\, dz \right] dx\, dy \\ &= \iint_D \left[R(x, y, f_2(x, y)) - R(x, y, f_1(x, y)) \right] dx\, dy.\end{aligned}$$

(f) Write out the boundary integral using the formulas for the surface element of the bounding graphs:

$$d\mathbf{S} = \left(-\frac{\partial f_2}{\partial x}\mathbf{i} - \frac{\partial f_2}{\partial y}\mathbf{j} + \mathbf{k} \right) dx\, dy,$$

and

$$d\mathbf{S} = \left(-\frac{\partial f_1}{\partial x}\mathbf{i} - \frac{\partial f_1}{\partial y}\mathbf{j} - \mathbf{k} \right) dx\, dy,$$

(g) Note that on the upper surface

$$R\mathbf{k} \cdot d\mathbf{S} = R(x, y, f_2(x, y))\, dx\, dy$$

while on the lower surface,

$$R\mathbf{k} \cdot d\mathbf{S} = -R(x, y, f_1(x, y))\, dx\, dy$$

(h) There is no contribution to the surface integral from the sides of the region as $R\mathbf{k}$ and $d\mathbf{S}$ are orthogonal. Comparing this with the preceding formula for the triple integral of $\partial R/\partial z$ gives the result.

4. As with Green's and Stokes' Theorems, the result is seen to be valid on a more general region, by breaking it up into a union of symmetric elementary regions.

5. From the divergence theorem and the mean value theorem, it follows that

$$\nabla \cdot \mathbf{F}(P) = \lim_{\rho \to 0} \frac{1}{V(W_\rho)} \iint_{\partial W_\rho} \mathbf{F} \cdot d\mathbf{S}$$

where W_ρ is a family of regions that approaches the point P as ρ tends to zero. This makes precise the idea (already discussed in Chapter 4) that the divergence is the net outward ***flux per unit volume.***

6. A vector field $\mathbf{F}$ is called ***divergence free*** or ***incompressible*** when $\nabla \cdot \mathbf{F} = 0$. By the divergence theorem this is equivalent to the property that the flux of $\mathbf{F}$ out of any surface is zero. This agrees with the earlier intuition about the divergence as the rate of change of volume under motion along flow lines.

7. ***Gauss' Law*** states that for a region W containing the origin,

$$\iint_{\partial W} \frac{\mathbf{r} \cdot d\mathbf{S}}{r^3} = 4\pi$$

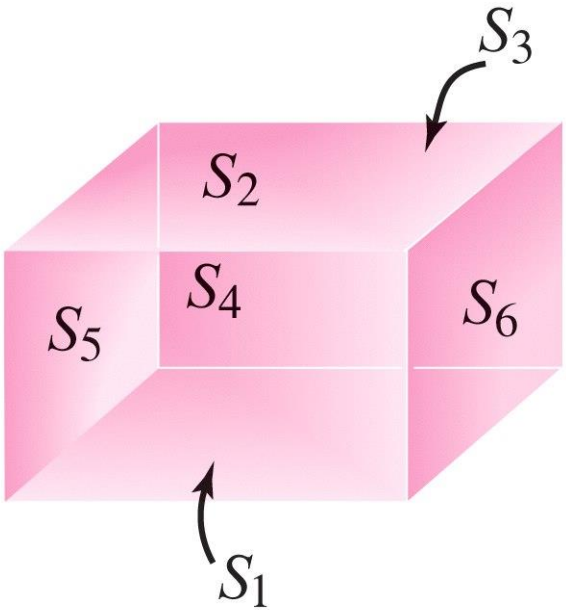
(the integral is zero if the region does not contain the origin). This is a good example where students must be a little careful with places where the integrand is not defined. One uses the divergence theorem to write

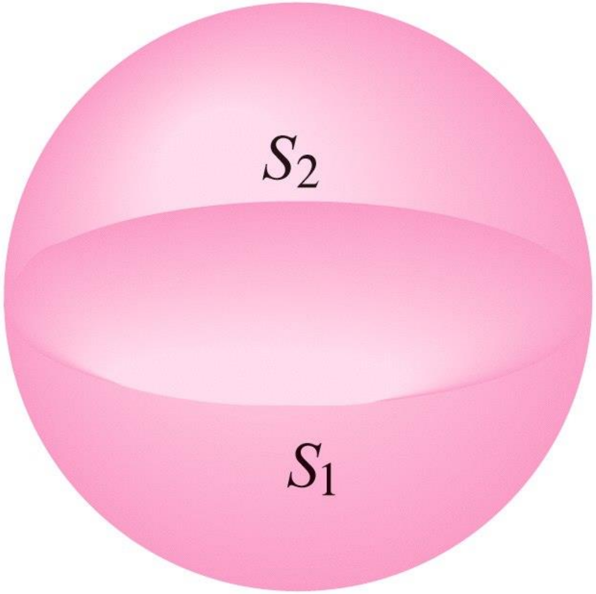
$$\iint_{\partial W} \frac{\mathbf{r} \cdot d\mathbf{S}}{r^3} = \iiint_W \nabla \cdot \frac{\mathbf{r}}{r^3}\, dV$$

but for $r \neq 0$, $\nabla \cdot (\mathbf{r}/r^3) = 0$. Thus, one can deform the region to that of a small sphere surrounding the origin and for the sphere one evaluates the integral easily to be 4π.

Closed elementary surfaces

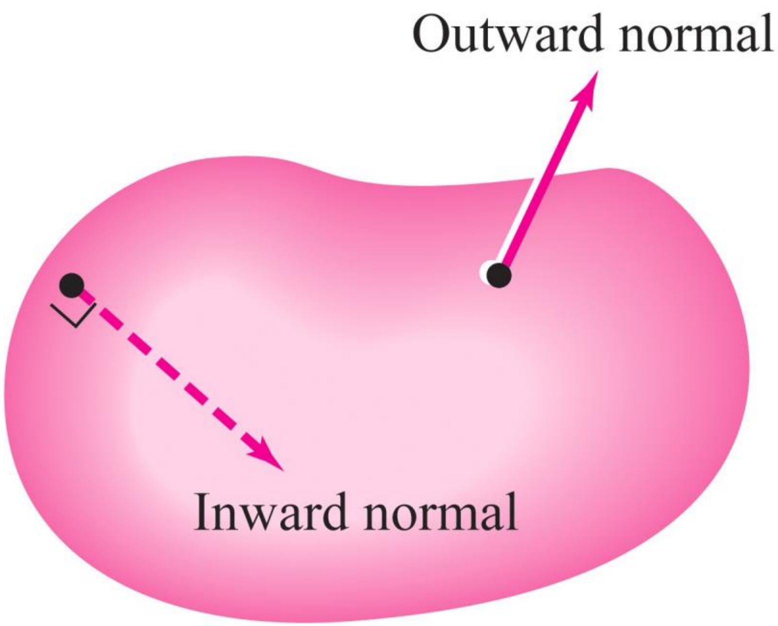
$$S = \bigcup_i S_i$$





$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \sum_i \iint_{S_i} \mathbf{F} \cdot d\mathbf{S}$$

S a closed surface enclosing a region W



$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S (\mathbf{F} \cdot \mathbf{n})\ dS$$

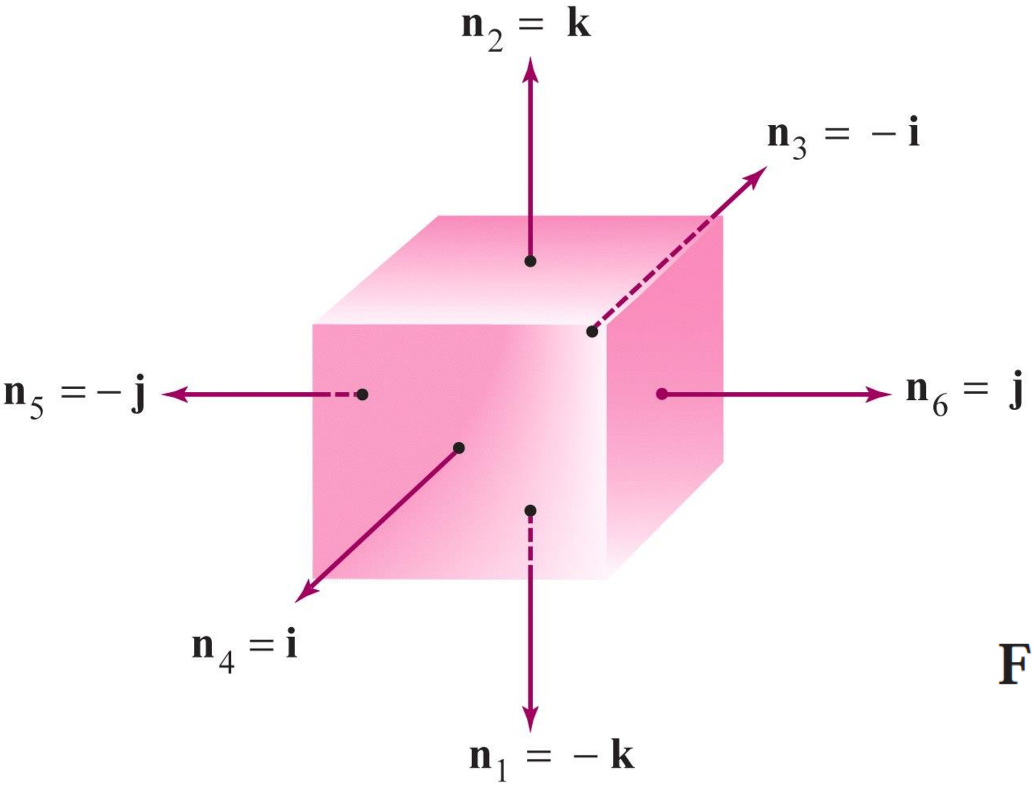
∂W enclosing W in "outward" orientation

∂W_{op} enclosing W in the opposite "inward" orientation

$$\iint_{\partial W} \mathbf{F} \cdot d\mathbf{S} = -\iint_{\partial W_{\mathrm{op}}} \mathbf{F} \cdot d\mathbf{S}$$

The unit cube W given by

$$0 \leq x \leq 1, \qquad 0 \leq y \leq 1, \qquad 0 \leq z \leq 1$$



$$\begin{aligned}
&S_1\colon z = 0, && 0 \leq x \leq 1, && 0 \leq y \leq 1\\
&S_2\colon z = 1, && 0 \leq x \leq 1, && 0 \leq y \leq 1\\
&S_3\colon x = 0, && 0 \leq y \leq 1, && 0 \leq z \leq 1\\
&S_4\colon x = 1, && 0 \leq y \leq 1, && 0 \leq z \leq 1\\
&S_5\colon y = 0, && 0 \leq x \leq 1, && 0 \leq z \leq 1\\
&S_6\colon y = 1, && 0 \leq x \leq 1, && 0 \leq z \leq 1
\end{aligned}$$

$$\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$$

$$\begin{aligned}
\iint_{\partial W} \mathbf{F}\cdot d\mathbf{S} = \iint_S \mathbf{F}\cdot\mathbf{n}\, dS = -\iint_{S_1} F_3\, dS + \iint_{S_2} F_3\, dS - \iint_{S_3} F_1\, dS\\
+ \iint_{S_4} F_1\, dS - \iint_{S_5} F_2\, dS + \iint_{S_6} F_2\, dS. \quad \blacktriangle
\end{aligned}$$

THEOREM 9: Gauss' Divergence Theorem Let W be a symmetric elementary region in space. Denote by ∂W the oriented closed surface that bounds W. Let $\mathbf{F}$ be a smooth vector field defined on W. Then

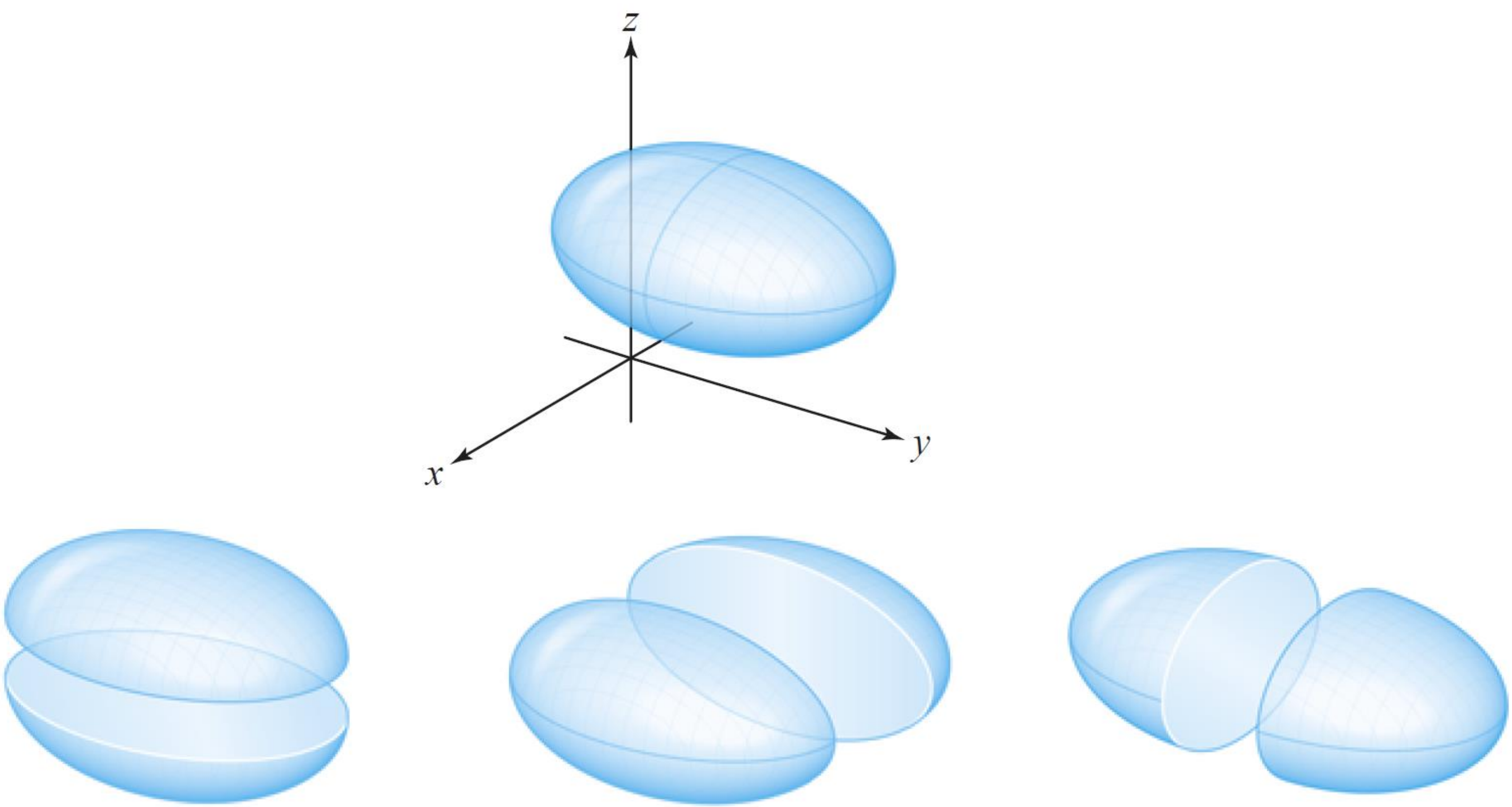
$$\iiint_W (\nabla \cdot \mathbf{F})\, dV = \iint_{\partial W} \mathbf{F} \cdot d\mathbf{S}$$

or, alternatively,

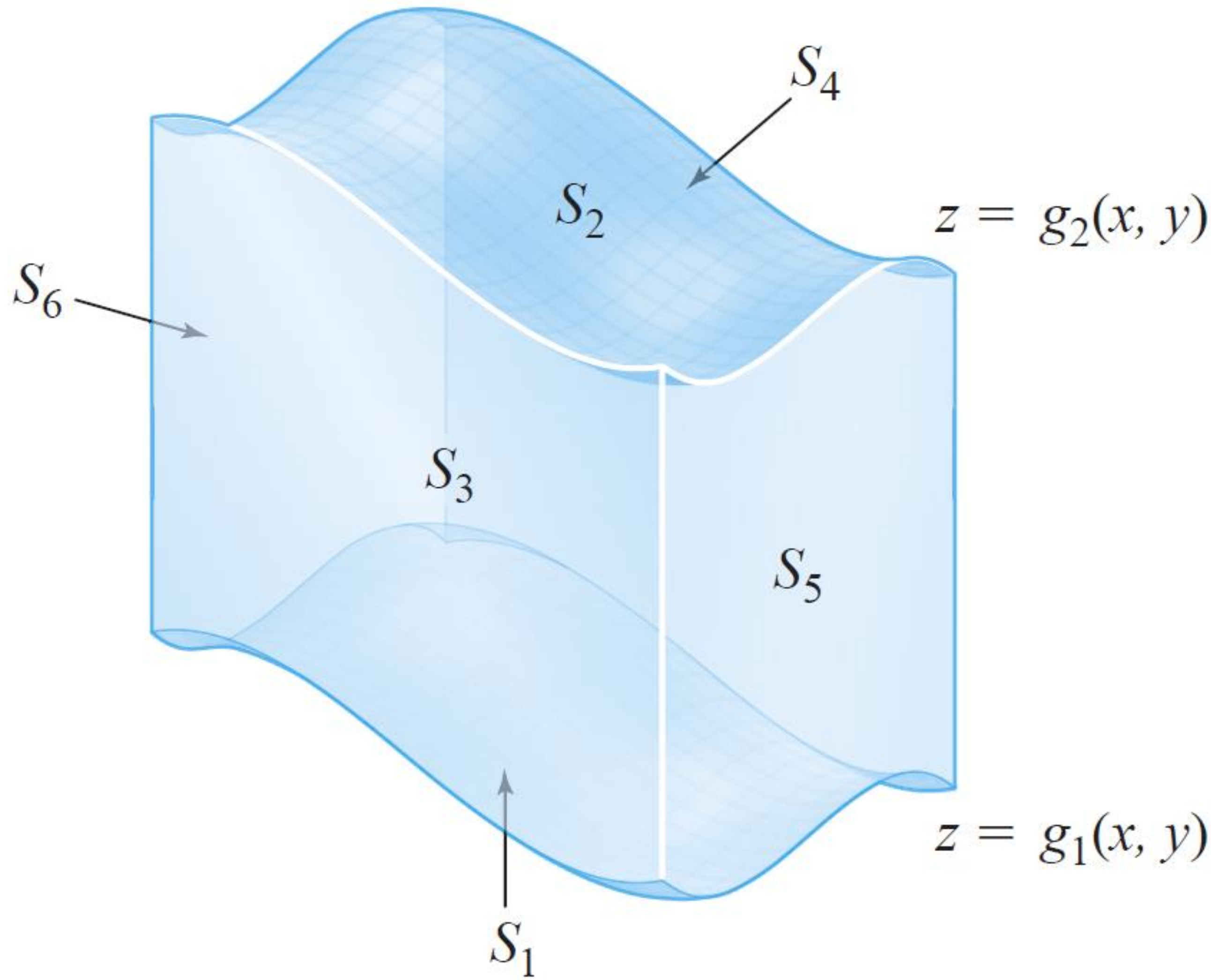
$$\iiint_W (\operatorname{div} \mathbf{F})\, dV = \iint_{\partial W} (\mathbf{F} \cdot \mathbf{n})\, dS.$$

Symmetric elementary regions

They are elementary regions simultaneously for the three axes



Elementary region with respect to the z-axis



proof If $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$, then by definition, the divergence of $\mathbf{F}$ is given by $\operatorname{div} \mathbf{F} = \partial P/\partial x + \partial Q/\partial y + \partial R/\partial z$, so we can write (using additivity of the volume integral)

$$\iiint_W \operatorname{div} \mathbf{F}\, dV = \iiint_W \frac{\partial P}{\partial x}\, dV + \iiint_W \frac{\partial Q}{\partial y}\, dV + \iiint_W \frac{\partial R}{\partial z}\, dV.$$

On the other hand, the surface integral in question is

$$\begin{aligned}\iint_{\partial W} \mathbf{F} \cdot \mathbf{n}\, dS &= \iint_{\partial W} (P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}) \cdot \mathbf{n}\, dS \\ &= \iint_{\partial W} P\mathbf{i} \cdot \mathbf{n}\, dS + \iint_{\partial W} Q\mathbf{j} \cdot \mathbf{n}\, dS + \iint_{\partial W} R\mathbf{k} \cdot \mathbf{n}\, dS.\end{aligned}$$

The theorem will follow if we establish the three equalities

$$\iint_{\partial W} P\mathbf{i} \cdot \mathbf{n}\, dS = \iiint_W \frac{\partial P}{\partial x}\, dV,$$

$$\iint_{\partial W} Q\mathbf{j} \cdot \mathbf{n}\, dS = \iiint_W \frac{\partial Q}{\partial y}\, dV,$$

$$\iint_{\partial W} R\mathbf{k} \cdot \mathbf{n}\, dS = \iiint_W \frac{\partial R}{\partial z}\, dV. \qquad (3)$$

We shall prove equation (3); the other two equalities can be proved in an analogous fashion.

Because W is a symmetric elementary region, there is a pair of functions

$$z = g_1(x, y), \qquad z = g_2(x, y),$$

with common domain an elementary region D in the xy plane, such that W is the set of all points (x, y, z) satisfying

$$g_1(x, y) \leq z \leq g_2(x, y), \qquad (x, y) \in D.$$

By reduction to iterated integrals, we have

$$\iiint_W \frac{\partial R}{\partial z}\, dV = \iint_D \left(\int_{z=g_1(x,y)}^{z=g_2(x,y)} \frac{\partial R}{\partial z}\, dz \right) dx\, dy,$$

and so, by the fundamental theorem of calculus,

$$\iiint_W \frac{\partial R}{\partial z}\, dV = \iint_D [R(x, y, g_2(x, y)) - R(x, y, g_1(x, y))]\, dx\, dy. \tag{4}$$

The boundary of W is a closed surface whose top S_2 is the graph of $z = g_2(x, y)$, where $(x, y) \in D$, and whose bottom S_1 is the graph of $z = g_1(x, y)$, $(x, y) \in D$. The four other sides of ∂W consist of surfaces S_3, S_4, S_5, and S_6, whose normals are always perpendicular to the z axis. (See Figure 8.4.4. Note that some of the other four sides might be absent; for instance, if W is a solid ball and ∂W is a sphere.) By definition,

$$\iint_{\partial W} R\mathbf{k} \cdot \mathbf{n}\, dS = \iint_{S_1} R\mathbf{k} \cdot \mathbf{n}_1\, dS + \iint_{S_2} R\mathbf{k} \cdot \mathbf{n}_2\, dS + \sum_{i=3}^{6} \iint_{S_i} R\mathbf{k} \cdot \mathbf{n}_i\, dS.$$

Because the normal $\mathbf{n}_i$ is perpendicular to $\mathbf{k}$ on each of S_3, S_4, S_5, S_6, we have $\mathbf{k} \cdot \mathbf{n} = 0$ along these faces, and so the integral reduces to

$$\iint_{\partial W} R\mathbf{k} \cdot \mathbf{n}\, ds = \iint_{S_1} R\mathbf{k} \cdot d\mathbf{S}_1 + \iint_{S_2} R\mathbf{k} \cdot d\mathbf{S}_2. \tag{5}$$

The surface S_1 is defined by $z = g_1(x, y)$, and

$$d\mathbf{S}_1 = \left(\frac{\partial g_1}{\partial x}\mathbf{i} + \frac{\partial g_1}{\partial y}\mathbf{j} - \mathbf{k} \right) dx\, dy$$

(the negative of the general formula for $d\mathbf{S}$ for graphs from Section 7.6, because the normal is downward pointing). Therefore,

$$\iint_{S_1} R\mathbf{k} \cdot d\mathbf{S}_1 = -\iint_D R(x, y, g_1(x, y))\; dx\, dy. \tag{6}$$

Similarly, for the top face S_2,

$$d\mathbf{S}_2 = \left(-\frac{\partial g_2}{\partial x}\mathbf{i} - \frac{\partial g_2}{\partial y}\mathbf{j} + \mathbf{k}\right) dx\, dy.$$

Therefore,

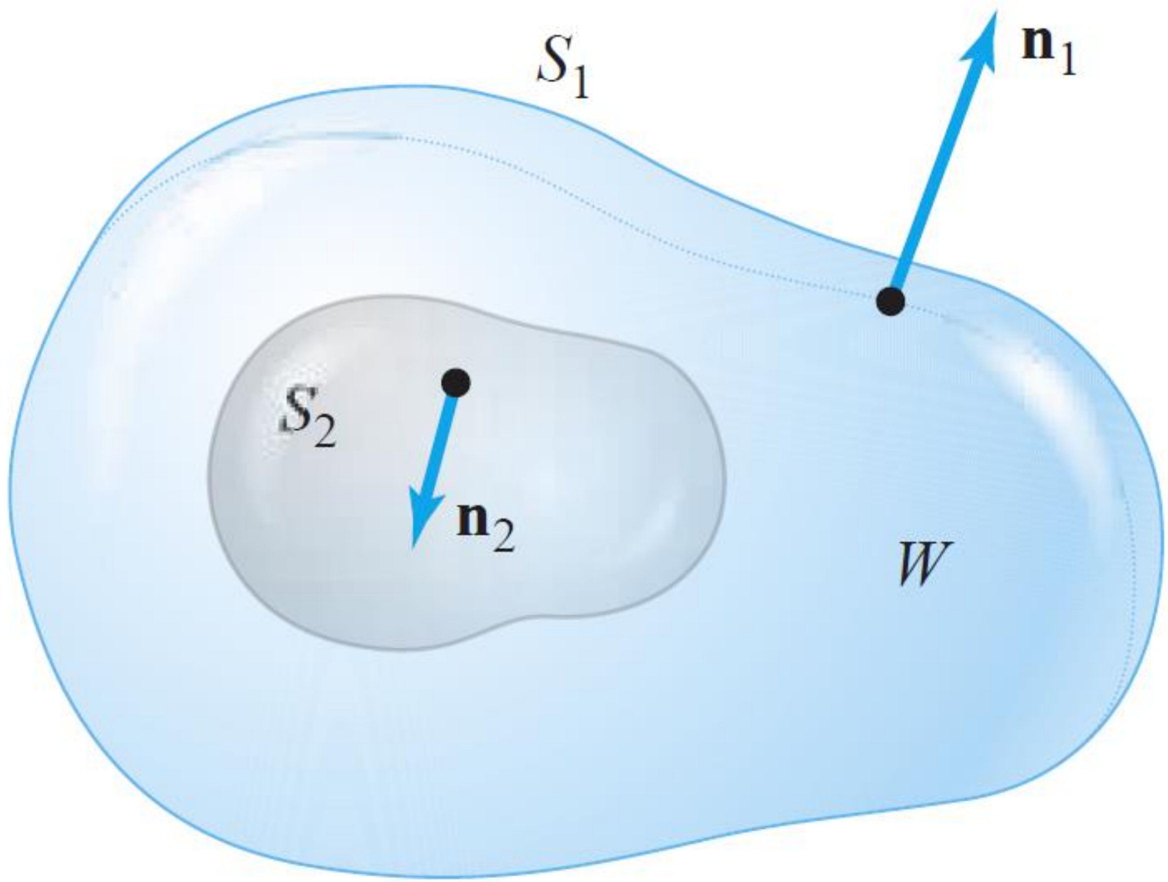
$$\iint_{S_2} R\mathbf{k} \cdot d\mathbf{S}_2 = \iint_D R(x, y, g_2(x, y))\; dx\, dy. \tag{7}$$

Substituting equations (6) and (7) into equation (5) and then comparing with equation (4), we obtain

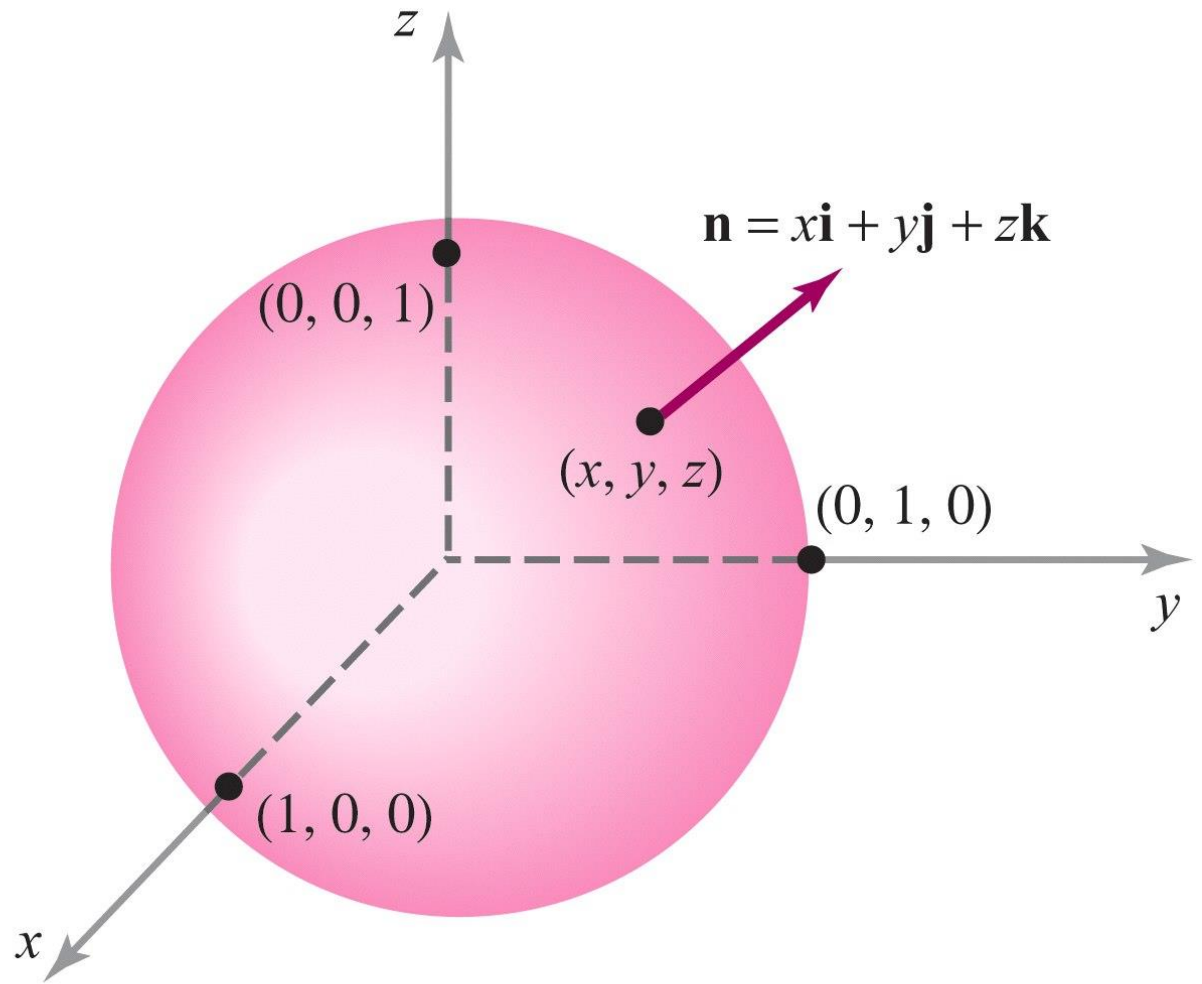
$$\iiint_W \frac{\partial R}{\partial z}\, dV = \iint_{\partial W} R(\mathbf{k} \cdot \mathbf{n})\; dS.$$

The remaining equalities, (1) and (2), can be established in the same way to complete the proof. ■

A more general region to which Gauss' Theorem applies



Consider $\mathbf{F} = 2x\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$. Let S be the unit sphere defined by $x^2 + y^2 + z^2 = 1$. Evaluate $\iint_S \mathbf{F} \cdot \mathbf{n}\, dS$.



Consider $\mathbf{F} = 2x\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$. Let S be the unit sphere defined by $x^2 + y^2 + z^2 = 1$. Evaluate $\iint_S \mathbf{F} \cdot \mathbf{n}\, dS$.

By Gauss' theorem,

$$\iint_S \mathbf{F} \cdot \mathbf{n}\, dS = \iiint_W (\operatorname{div} \mathbf{F})\, dV,$$

where W is the ball bounded by the sphere. The integral on the right is

$$2 \iiint_W (1 + y + z)\, dV = 2 \iiint_W dV + 2 \iiint_W y\, dV + 2 \iiint_W z\, dV.$$

By symmetry, we can argue that $\iiint_W y\, dV = \iiint_W z\, dV = 0$ (see Exercise 21, Section 6.3). Thus, because a sphere of radius R has volume $4\pi R^3/3$,

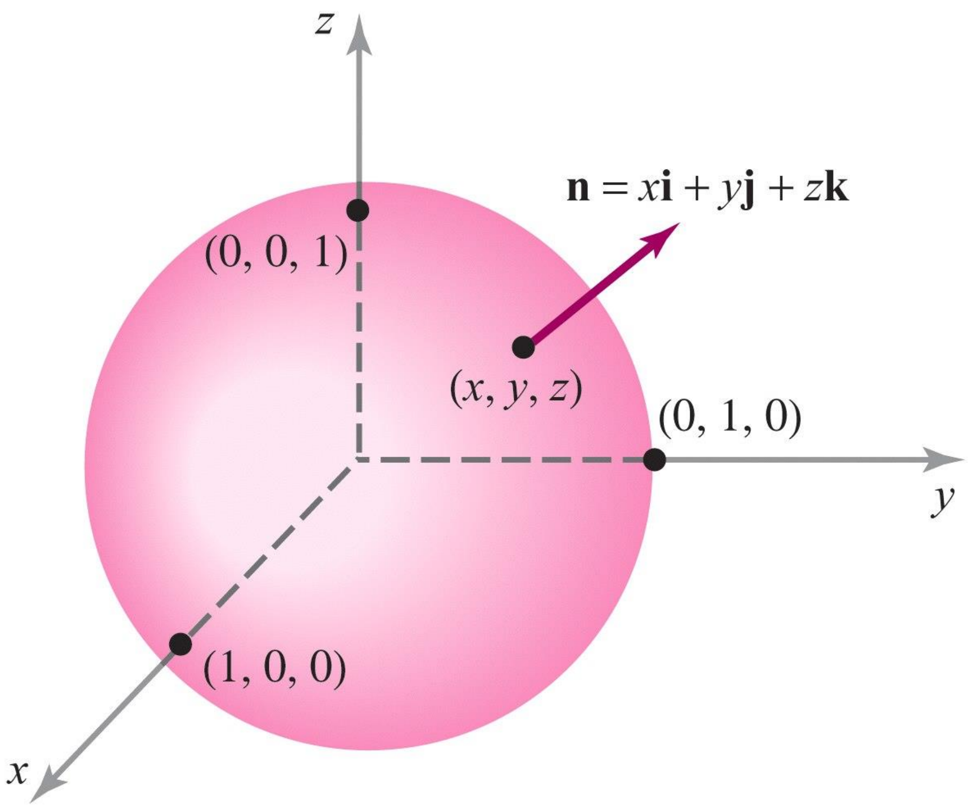
$$2 \iiint_W (1 + y + z)\, dV = 2 \iiint_W dV = \frac{8\pi}{3}.$$

You can convince yourself that direct computation of $\iint_S \mathbf{F} \cdot \mathbf{n}\, dS$ proves unwieldy. ▲

Use the divergence theorem to evaluate

$$\iint_{\partial W} (x^2 + y + z)\, dS,$$

where W is the solid ball $x^2 + y^2 + z^2 \leq 1$.



To apply Gauss' divergence theorem, we find a vector field $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$ on W with $\mathbf{F} \cdot \mathbf{n} = x^2 + y + z$. At any point $(x, y, z) \in \partial W$, the outward unit normal $\mathbf{n}$ to ∂W is

$$\mathbf{n} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k},$$

because on ∂W, $x^2 + y^2 + z^2 = 1$ and the radius vector $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ is normal to the sphere ∂W (Figure 8.4.6).

Therefore, if $\mathbf{F}$ is the desired vector field, then

$$\mathbf{F} \cdot \mathbf{n} = F_1 x + F_2 y + F_2 z.$$

We set

$$F_1 x = x^2, \qquad F_2 y = y, \qquad F_3 z = z$$

and solve for F_1, F_2, and F_3 to find that $\mathbf{F} = x\mathbf{i} + \mathbf{j} + \mathbf{k}$. Computing div $\mathbf{F}$, we get

$$\operatorname{div} \mathbf{F} = 1 + 0 + 0 = 1.$$

Thus, by Gauss' divergence theorem,

$$\iint_{\partial W} (x^2 + y + z)\, dS = \iiint_W dV = \text{volume } (W) = \frac{4}{3}\pi.$$

Evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = xy^2\mathbf{i} + x^2y\mathbf{j} + y\mathbf{k}$ and S is the surface of the cylinder $x^2 + y^2 = 1$, bounded by the planes $z = 1$ and $z = -1$, and including the portions $x^2 + y^2 \leq 1$ when $z = \pm 1$.

Evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = xy^2\mathbf{i} + x^2y\mathbf{j} + y\mathbf{k}$ and S is the surface of the cylinder $x^2 + y^2 = 1$, bounded by the planes $z = 1$ and $z = -1$, and including the portions $x^2 + y^2 \leq 1$ when $z = \pm 1$.

We can compute this integral directly, but it is easier to use the divergence theorem.

Now S is the boundary of the region W given by $x^2 + y^2 \leq 1$, $-1 \leq z \leq 1$. Thus, $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_W (\operatorname{div} \mathbf{F})\, dV$. Moreover,

$$\iiint_W (\operatorname{div} \mathbf{F})\, dV = \iiint_W (x^2 + y^2)\ dx\,dy\,dz = \int_{-1}^{1} \left(\iint_{x^2+y^2 \leq 1} (x^2 + y^2)\ dx\,dy \right) dz$$
$$= 2 \iint_{x^2+y^2 \leq 1} (x^2 + y^2)\ dx\,dy.$$

We change variables to polar coordinates to evaluate the double integral:

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad 0 \leq r \leq 1, \qquad 0 \leq \theta \leq 2\pi.$$

Hence, we have $\partial(x, y)/\partial(r, \theta) = r$ and $x^2 + y^2 = r^2$. Thus,

$$\iint_{x^2+y^2 \leq 1} (x^2 + y^2)\ dx\,dy = \int_0^{2\pi} \left(\int_0^1 r^3\, dr \right) d\theta = \frac{1}{2}\pi.$$

Therefore, $\iiint_W \operatorname{div} \mathbf{F}\, dV = \pi$. ▲

The Divergence as the Flux per Unit Volume

The physical meaning of divergence is that at a point P, div $\mathbf{F}(\mathrm{P})$ is the rate of net outward flux at P per unit volume. This follows from Gauss' theorem and the mean-value theorem for integrals: If W_ρ is a ball in $\mathbb{R}^3$ of radius ρ centered at P, then there is a point $\mathrm{Q} \in W_\rho$ such that

$$\iint_{\partial W_\rho} \mathbf{F}\cdot\mathbf{n}\,dS = \iiint_{W_\rho} \operatorname{div}\mathbf{F}\,dV = \operatorname{div}\mathbf{F}(\mathrm{Q})\cdot \text{volume}\,(W_\rho)$$

and so

$$\operatorname{div}\mathbf{F}(\mathrm{P}) = \underset{\rho\to 0}{\text{limit}}\operatorname{div}\mathbf{F}(\mathrm{Q}) = \underset{\rho\to 0}{\text{limit}}\frac{1}{V(W_\rho)}\iint_{\partial W_\rho} \mathbf{F}\cdot\mathbf{n}\,dS.$$

This is analogous to the limit formulation of the curl given at the end of Section 8.2. Thus, if div $\mathbf{F}(\mathrm{P}) > 0$, we consider P to be a ***source***, for there is a net outward flow near P. If div $\mathbf{F}(\mathrm{P}) < 0$, P is called a ***sink*** for $\mathbf{F}$.

A C^1 vector field $\mathbf{F}$ defined on $\mathbb{R}^3$ is said to be ***divergence-free*** if div $\mathbf{F} = 0$.
A fluid with this property is therefore described as ***incompressible***.

THEOREM 10: Gauss' Law Let M be a symmetric elementary region in $\mathbb{R}^3$. Then if $(0, 0, 0) \notin \partial M$, we have

$$\iint_{\partial M} \frac{\mathbf{r} \cdot \mathbf{n}}{r^3}\, dS = \begin{cases} 4\pi & \text{if } (0, 0, 0) \in M \\ 0 & \text{if } (0, 0, 0) \notin M, \end{cases}$$

where

$$\mathbf{r}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

and

$$r(x, y, z) = \|\mathbf{r}(x, y, z)\| = \sqrt{x^2 + y^2 + z^2}.$$

proof of Gauss' law First suppose $(0, 0, 0) \notin M$. Then $\mathbf{r}/r^3$ is a C^1 vector field on M and ∂M, and so by the divergence theorem,

$$\iint_{\partial M} \frac{\mathbf{r} \cdot \mathbf{n}}{r^3}\, dS = \iiint_M \nabla \cdot \left(\frac{\mathbf{r}}{r^3}\right) dV.$$

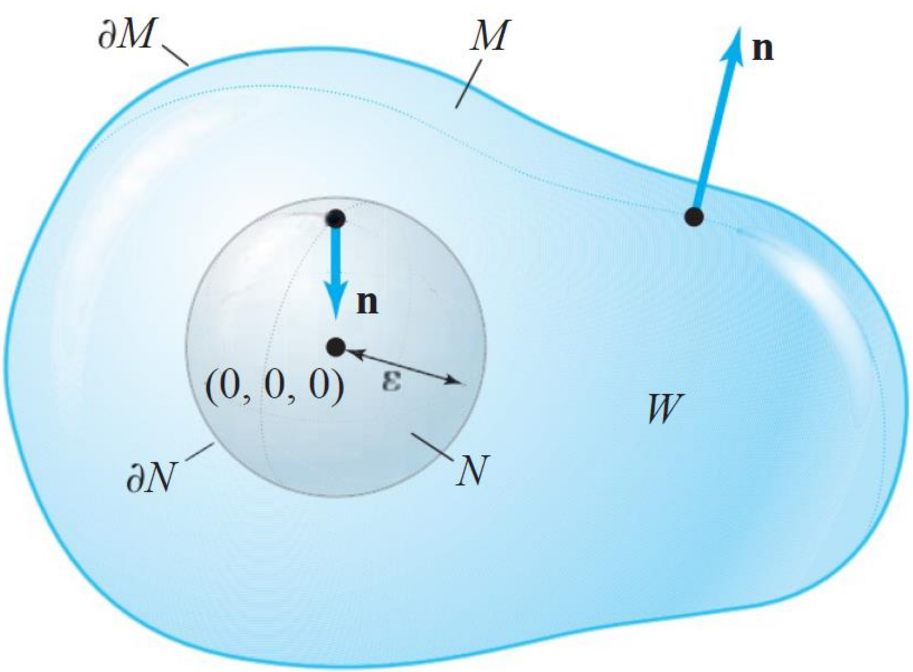
But $\nabla \cdot (\mathbf{r}/r^3) = 0$ for $r \neq 0$, as you can easily verify (see Exercise 38, Section 4.4). Thus,

$$\iint_{\partial M} \frac{\mathbf{r} \cdot \mathbf{n}}{r^3}\, dS = 0.$$

Now let us suppose $(0, 0, 0) \in M$. We can no longer use the preceding method, because $\mathbf{r}/r^3$ is not smooth on M, due to the zero denominator at $\mathbf{r} = (0, 0, 0)$. Because $(0, 0, 0) \in M$ and $(0, 0, 0) \notin \partial M$, there is an $\varepsilon > 0$ such that the ball N of radius ε centered at $(0, 0, 0)$ is contained completely inside M. Let W be the region between M and N. Then W has boundary $\partial N \cup \partial M = S$. But the orientation on ∂N induced by the *outward* normal on W is the opposite of that obtained from N (see Figure 8.4.7).

Now $\nabla \cdot (\mathbf{r}/r^3) = 0$ on W, and so, by the divergence theorem applied to the (nonelementary) region W,

$$\iint_S \frac{\mathbf{r} \cdot \mathbf{n}}{r^3}\, dS = \iiint_W \nabla \cdot \left(\frac{\mathbf{r}}{r^3}\right) dV = 0.$$



Because

$$\iint_S \frac{\mathbf{r}\cdot\mathbf{n}}{r^3}\,dS = \iint_{\partial M} \frac{\mathbf{r}\cdot\mathbf{n}}{r^3}\,dS + \iint_{\partial N} \frac{\mathbf{r}\cdot\mathbf{n}}{r^3}\,dS,$$

where $\mathbf{n}$ is the outward normal to S, we have

$$\iint_{\partial M} \frac{\mathbf{r}\cdot\mathbf{n}}{r^3}\,dS = -\iint_{\partial N} \frac{\mathbf{r}\cdot\mathbf{n}}{r^3}\,dS.$$

However, on ∂N, $\mathbf{n} = -\mathbf{r}/r$ and $r = \varepsilon$, because ∂N is a sphere of radius ε, so that

$$-\iint_{\partial N} \frac{\mathbf{r}\cdot\mathbf{n}}{r^3}\,dS = \iint_{\partial N} \frac{\varepsilon^2}{\varepsilon^4}\,dS = \frac{1}{\varepsilon^2}\iint_{\partial N} dS.$$

But $\iint_{\partial N} dS = 4\pi\varepsilon^2$, the surface area of the sphere of radius ε. This proves the result. ■

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Vector Calculus

Fifth Edition

Chapter 8:
The Integral Theorems of Vector Analysis

8.5 Some Differential Equations of Mechanics and Technology

8.5 Applications: Physics, Engineering & Differential Equations

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Key Points in this Section.

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1. The ***law of conservation of mass*** for a vector field $\mathbf{V}$ and a function ρ, is the condition

$$\frac{d}{dt}\iiint_W \rho\, dV = -\iint_{\partial W} \mathbf{J}\cdot\mathbf{n}\, dA$$

where $\mathbf{J} = \rho\mathbf{V}$ and where W is an arbitrary region in $\mathbb{R}^3$.

2. The divergence theorem shows that conservation of mass is equivalent to the ***continuity equation***

$$\operatorname{div}\mathbf{J} + \frac{\partial \rho}{\partial t} = 0.$$

3. The ***material derivative*** of a function f with respect to a vector field $\mathbf{F}$ is

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + \nabla f\cdot\mathbf{F}.$$

4. If $\phi(x,t)$ is the ***flow*** of the vector field $\mathbf{F}$, (that is, $\phi(x,0) = x$ and the map $t \mapsto \phi(x,t)$ for each fixed x is a flow line of F), and J is the Jacobian determinant of the flow map $x \mapsto \phi(x,t)$, then

$$\frac{\partial J}{\partial t} = J \operatorname{div} \mathbf{F}$$

and the ***transport theorem*** holds for any function f of (x,y,z,t):

$$\frac{d}{dt} \iiint_{W_t} f \, dV = \iiint_{W_t} \left(\frac{Df}{Dt} + f \operatorname{div} \mathbf{F} \right) dV$$

where W_t is the image of a region W in $\mathbb{R}^3$ under the flow map.

5. ***Euler's equation for a perfect fluid*** is

$$\rho \left(\frac{\partial \mathbf{V}}{\partial t} + \mathbf{V} \cdot \nabla \mathbf{V} \right) = -\nabla p$$

where $\mathbf{V}$ is the fluid velocity field, ρ is the fluid density and p is the pressure.

6. Conservation of energy applied to heat energy gives the ***heat equation***:

$$\frac{\partial T}{\partial t} = k\nabla^2 T,$$

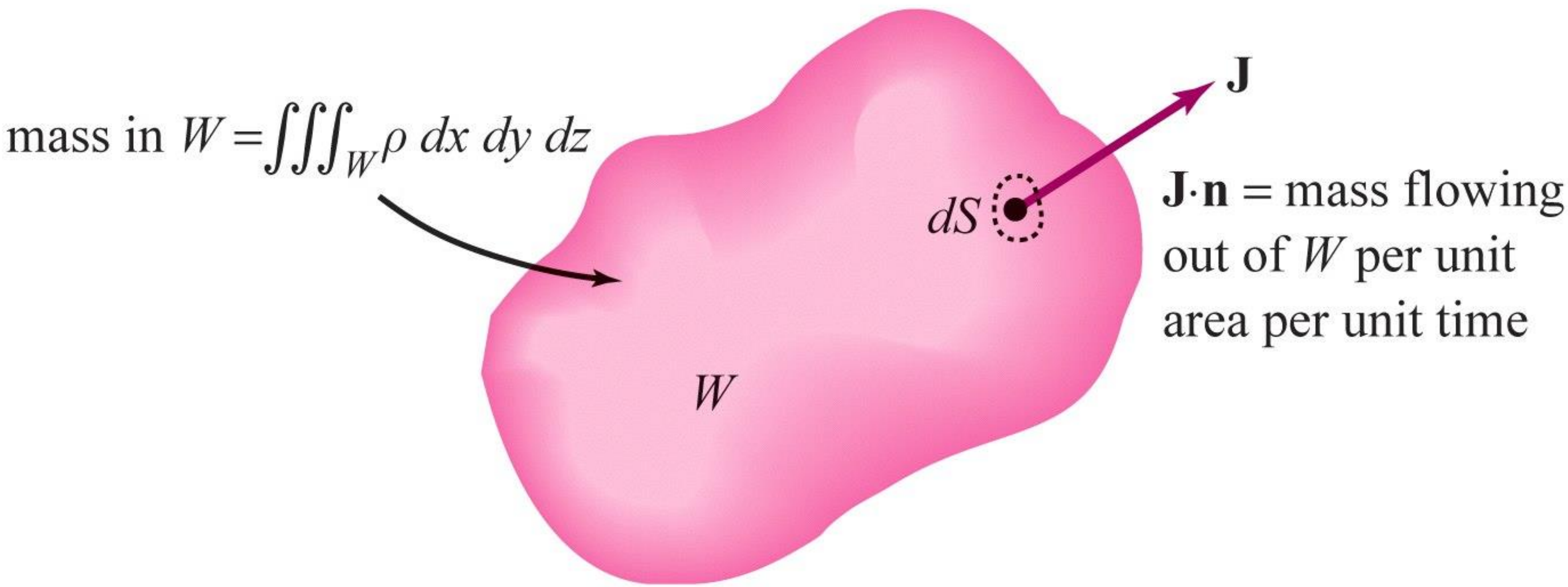
where T is the temperature and k is the material heat conductivity.

7. ***Maxwell's equations*** for an electric field $\mathbf{E}$ and a magnetic field $\mathbf{H}$ state that

$$\begin{aligned}
\operatorname{div} \mathbf{E} &= \rho \\
\operatorname{div} \mathbf{H} &= 0 \\
\operatorname{curl} \mathbf{E} + \frac{\partial \mathbf{H}}{\partial t} &= 0 \\
\operatorname{curl} \mathbf{H} - \frac{\partial \mathbf{E}}{\partial t} &= J,
\end{aligned}$$

where ρ is the charge density and J is the current.

8. Stokes' and Gauss' theorems are the key to understanding the integral versions of these equations. For example, the integral version of the last of Maxwell's equations is ***Faraday's law***, which was studied in §8.2 (see Example 5).



mass in $W = \iiint_W \rho \, dx \, dy \, dz$
W
dS
$\mathbf{J}$
$\mathbf{J} \cdot \mathbf{n}$ = mass flowing out of W per unit area per unit time

THEOREM 11 For $\mathbf{V}$ and ρ (a smooth vector field and a scalar field on $\mathbb{R}^3$), the law of conservation of mass for $\mathbf{V}$ and ρ is equivalent to the condition

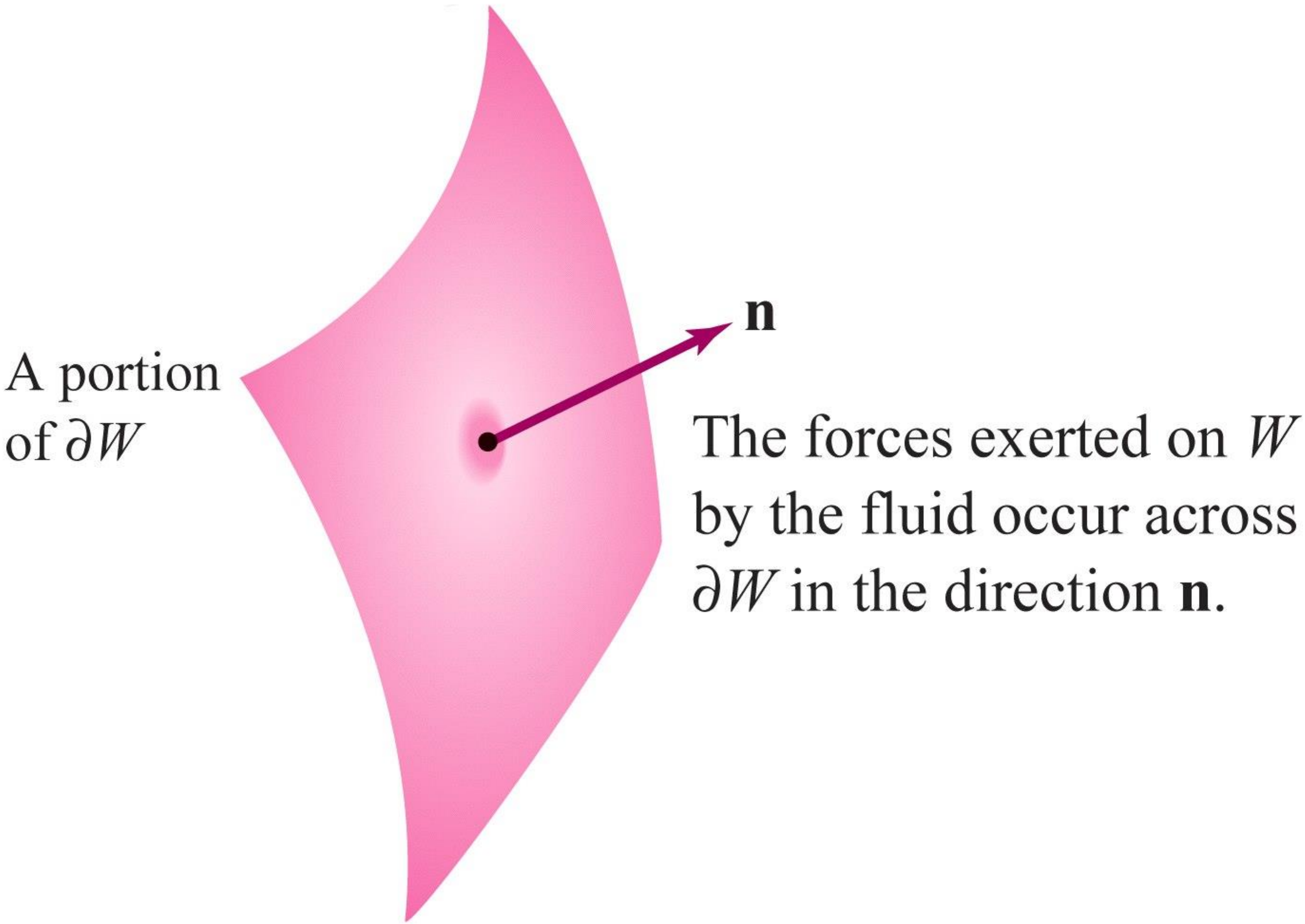
$$\operatorname{div} \mathbf{J} + \frac{\partial \rho}{\partial t} = 0. \tag{1}$$

That is,

$$\rho \operatorname{div} \mathbf{V} + \mathbf{V} \cdot \nabla \rho + \frac{\partial \rho}{\partial t} = 0. \tag{1'}$$

THEOREM 12 Let $\mathbf{F}$ be a vector field on $\mathbb{R}^3$ and denote the flow line of $\mathbf{F}$ starting at $\mathbf{x}$ after time t by $\phi(\mathbf{x}, t)$. (See the Internet supplement to Section 4.4 for more information.) Let $J(\mathbf{x}, t)$ be the Jacobian of the map $\phi_t \colon \mathbf{x} \mapsto \phi(\mathbf{x}, t)$ for t fixed. Then

$$\frac{\partial}{\partial t} J(\mathbf{x}, t) = [\operatorname{div} \mathbf{F}(\phi(\mathbf{x}, t))] J(\mathbf{x}, t).$$


n
A portion
of ∂W
The forces exerted on W
by the fluid occur across
∂W in the direction n.

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Chapter 8:
The Integral Theorems of Vector Analysis

8.6 Differential Forms

THEOREM 16: General Stokes' Theorem Let M be an oriented k-manifold in $\mathbb{R}^3$ ($k = 2$ or 3) contained in some open set K. Suppose ω is $(k-1)$-form on K. Then

$$\int_{\partial M} \omega = \int_M d\omega.$$